AUTO ELECTRONIC PARTS BOX

ELECTRICAL SCHEMATIC & WIRING DRAWING SET

แบบวงจรไฟฟ้าและการเดินสาย

AUTO-Parts-E01 | Rev. 4.0 | 2026-09-26 | DRAFT FOR REVIEW

PROJECT DATA

Mains supply	1/N/PE AC 220-230 V 50 Hz, 16 A
Earthing	TN-S, PE busbar -PE
Motor supply	60 V DC (drives X/Y), 24 V DC (drive Z)
Control voltage	24 V DC (PELV)
Standards	EN 60204-1, IEC 61082-1, IEC 81346-2, IEC 60617, IEC 60757
Axes	X belt (HBS860H), Y ball screw (HBS860H), Z lead screw (DM542)

REVISION HISTORY

Rev	Date	Description	By
3.0	2026-xx-xx	As-built sheets 01-07 (AI generated, block style)	-
4.0	2026-09-26	Complete redraw to IEC 61082 / 81346; safety circuit, fusing, terminal plan, wire list and BOM added	PAPON

HOW TO READ THIS DRAWING SET

Device tag -K1 = safety contactor. Letters per IEC 81346-2 (see sheet 02).

Wire number 503 = wire 03 of sheet 05. Every wire is listed in the wire list with colour, size and ends.

Cross-ref /05.3 = sheet 05, column 3. Arrows mark where a potential continues on another sheet.

Potentials L+1 / L+2 / L+P = 24 V DC circuits, M = 0 V, +60V / 0V60 = motor supply.

Terminals -X2:7 = terminal 7 of strip -X2. Terminal plan on sheet 14.

PAPON

DRAFT FOR REVIEW

PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

SHEET INDEX / สารบัญ

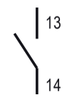
No.	Title	ชื่อแผ่น
01	Cover sheet, index, revisions	ปก สารบัญ ประวัติการแก้ไข
02	Legend & conventions	สัญลักษณ์และข้อกำหนด
03	System block diagram	ภาพรวมระบบ
04	AC mains & power supplies	ไฟ AC และแหล่งจ่ายไฟ
05	DC power distribution 60 V / 24 V	จ่ายไฟ DC
06	Safety circuit E-stop / -K1	วงจรความปลอดภัย
07	Main controller -A1 power & I/O	คอนโทรลเลอร์หลัก
08	Communication & peripherals	การสื่อสารและอุปกรณ์ต่อพ่วง
09	Motion MCU -A3 & level shifter -A4	ไมโครคอนโทรลเลอร์และบอร์ด NMOS
10	Stepper drives X / Y	ไดรเวอร์แกน X / Y
11	Stepper drive Z (dispenser)	ไดรเวอร์แกน Z
12	Remote I/O -A2 digital inputs	อินพุตดิจิทัล
13	Remote I/O -A2 digital outputs	เอาต์พุตดิจิทัล
14	Terminal plan -X1 / -X2 / -X3	ผังเทอร์มินอล
15	Engineering notes & open points	หมายเหตุและจุดที่ต้องตรวจสอบ
16	PARTS LIST (BOM) (1/2)	รายการอุปกรณ์
17	PARTS LIST (BOM) (2/2)	รายการอุปกรณ์
18	CABLE LIST	รายการสายเคเบิล
19	I/O LIST	รายการอินพุต / เอาต์พุต
20	Wire list (1/3)	รายการสายไฟ
21	Wire list (2/3)	รายการสายไฟ
22	Wire list (3/3)	รายการสายไฟ

COVER SHEET

ปกและสารบัญ

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DATE **2026-09-26**	APPROVED		

GRAPHICAL SYMBOLS (IEC 60617)



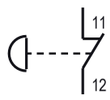
NO contact



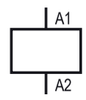
NC contact



Circuit breaker



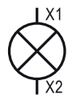
E-stop NC, latching



Relay / contactor coil



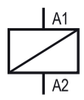
Fuse



Pilot light



Buzzer



Solenoid



Diode



Terminal



Protective earth



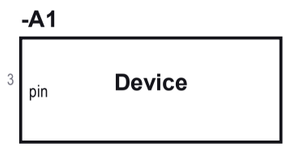
Stepper motor



Junction



Continues on sheet



Device (black box)



Partial / other sheet

Wire with number

WIRE COLOURS (EN 60204-1 13.2 / IEC 60757)

Code	Colour	Use
BN	Brown	AC line conductor L
BU	Blue (light)	AC neutral conductor N
GNYE	Green/Yellow	Protective earth PE
RD	Red	+60 V DC motor supply
BK	Black	0 V of 60 V motor supply
DBU	Dark blue	+24 V DC control circuits and 24 V signals
DBU/WH	Dark blue / white	0 V (M) of 24 V control supply
OG	Orange	Reserved: circuits not switched off by -Q1
(maker)	Manufacturer cable	Motor, encoder, sensor and USB/LAN cables

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

LEGEND

สัญ

503

with number

DEVICE LETTER CODES (IEC 81346-2)

Code	Class	Used for
A	Two or more purposes	Controllers, I/O, HMI, boards
B	Sensing -> signal	Proximity / photoelectric sensors
F	Protection	Fuses, SPD
K	Processing / switching signal	Contactor -K1
M	Mechanical energy (drive)	Stepper motors
P	Presenting information	Pilot lights, buzzer, speaker
Q	Switching energy flow	MCB -Q1
S	Manual operation -> signal	E-stop -S0
T	Converting energy	Power supplies
U	Converting (power electronics)	Stepper drives
V	Restricting flow	Diodes / suppressors
W	Guiding	Cables
X	Connecting	Terminals, inlet
Y	Mechanical actuation	Solenoid
Z	Filtering	EMI filter

POTENTIALS

Name	Value	Source / protection
L / N / PE	230 V AC	-X0, -Q1 C16, -Z1 -> -X1
+60V / 0V60	60 V DC	-T1 -> -K1:1-4 -> -F21/-F22 -> -U1/-U2
L+1	24 V DC	-T2 -> -F31 4 A -> -X2:1-4 (controllers, sensors, lamps)
L+2	24 V DC	-T2 -> -F34 2 A -> -X2:5-6 (safety circuit)
L+P	24 V DC	-T2 -> -F32 1 A -> -X2:7-8 (drive opto-couplers)
+24V Z	24 V DC	-T2 -> -F33 4 A -> -K1:5-6 -> -U3
M	0 V	-T2 0 V -> -X2:9-14, earthed at -X2:9
+5.1V	5.1 V DC	-T3 -> USB-C -> -A1

GENERAL WIRING RULES

1. All stranded conductors end in insulated ferrules (DIN 46228-4).
2. Wire markers at both ends showing the wire number.
3. Separate ducts: AC / 60 V power vs. 24 V, signal and LAN (>= 100 mm).
4. Shields of motor, encoder and sensor cables: 360 deg clamp to PE at entry.
5. Max. two conductors per terminal (twin ferrule).
6. Tightening torque per manufacturer; pull test on every crimp.

EGEND & CONVENTIONS

สัญลักษณ์และข้อกำหนด

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1 | 2 | 3 | 4

AC SUPPLY
-X0 inlet C14 + fuse
-Q1 MCB 2P C16
-F1 SPD type 2
-Z1 EMI filter -> -X1

-T1
PSU 60 V DC 400 W
S-400-60

60 V

-K1
SAFE

24 V

-T2
PSU 24 V DC 120 W
NDR-120-24

-T3
PSU 5.1 V DC 15 W
USB-C

DO0 enable

5.1 V

24 V

-A4
NMOS
3.

-A5
QR SCANNER
USB

USB1

-A6
CAMERA
USB UVC

USB2

-A1
MAIN CONTROLLER
Cytron IRIV PiControl CM4
Safety daemon, HMI server
Modbus TCP master
4 DI / 4 DO

-A7
AUDIO + SPEAKER
USB + -P5

USB3

USB

-A3

-A8
HMI KIOSK
10.1 in touch

eth0

Ethernet (OT)

-B1..-B10
FIELD SENSORS
8x inductive M12 PNP
2x Omron E3Z-D81
limits, home, drop, pickup

DI0-9

-A2
REMOTE I/O
Cytron IRIV IO
Modbus TCP 10.0.0.10
11 DI / 4 DO

DO0-3

-P

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AUTO ELECTRONIC PARTS BOX
Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE
SYSTEM

5 | 6 | 7 | 8

1

SAFETY CONTACTOR
4-pole DC
+ E-stop -S0

aux 21-22 -> -A2 DI10

-U1

DRIVE X
HBS860H

-M1

MOTOR X
NEMA34 8.5 Nm + enc

-U2

DRIVE Y
HBS860H

-M2

MOTOR Y
NEMA34 8.5 Nm + enc

-U3

DRIVE Z
DM542

-M3

ACTUATOR Z
NEMA17 T8x8

4

NMOS LEVEL SHIFTER
3.3 V -> 24 V open drain

STEP / DIR

3

MOTION MCU
NUCLEO-G491RE

Power

Control / hard-wired signal

Communication / pulse

SAFETY CONCEPT

E-stop -S0 opens the -K1 coil directly.
-K1 removes 60 V from -U1/-U2 and 24 V from -U3.
Controllers stay powered for diagnostics.
Software watchdog (Modbus > 350 ms) drops DO0.

-P1..-P4, -Y1

INDICATION & GATE
Pilot lights RD/YE/GN
Buzzer
Dispense solenoid

YSTEM BLOCK DIAGRAM

ภาพรวมระบบ

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1	2	3	4

-X0

IEC C14 inlet
fuse 10 A T

L N PE

-W401 3G1.5 H05VV-F
mains cord 230 V 50 Hz

401 402 403

-Q1
C16 2P
6 kA

1 3 2 4

404 405 406 407

-F1

L N

SPD
type 2

PE

-Z1

L N

EMI
20 A

PE 409

L' N'

408

CW4L2-20A-S

410 411

-PE cabinet / door bonding

X1:1 X1:4

412

-F11
6.3 A T

1 2

413 414 415

-T1

L N PE

PSU 60 V DC
6.7 A 400 W
S-400-60

+V -V

+V /05.1

-V /05.1

NOTES

Supply 1/N/PE
-F1 surge prote
Each power su
Connected load
Before -Q1 (ins

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AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

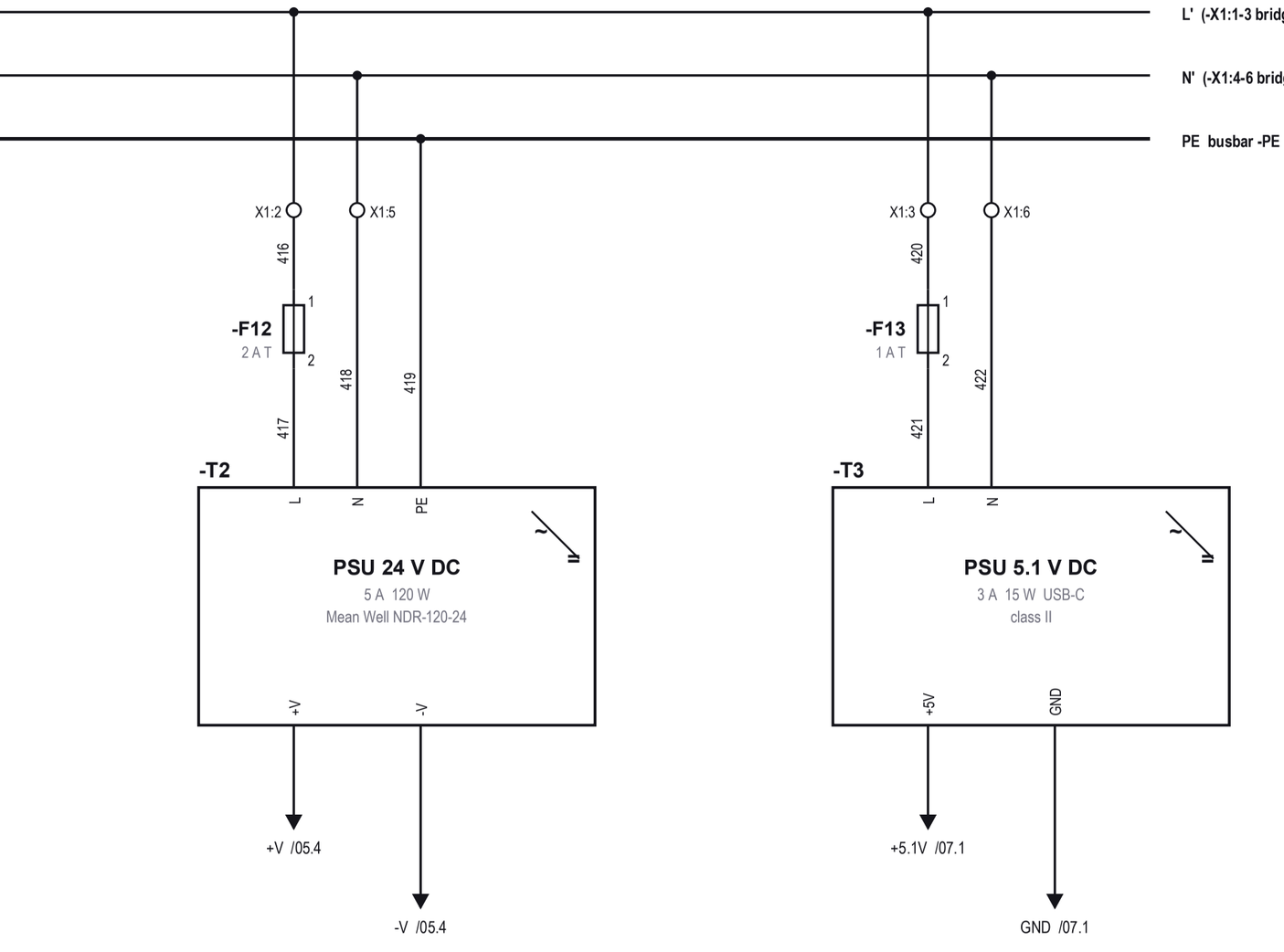
AC MAINS

ไฟ

	5	6	7	8

NOTES

1. Supply 1/N/PE AC 220-230 V 50 Hz, TN-S. Main switch and protection -Q1 (C16, 6 kA).
2. -F1 surge protection in parallel after -Q1; PE conductor 4 mm2, < 0.5 m.
3. Each power supply has its own fuse (-F11..-F13) for selectivity.
4. Connected load approx. 540 W -> I approx. 2.9 A at 230 V (PF 0.8).
5. Before -Q1 (inside -X0) the circuit stays live: keep covered.



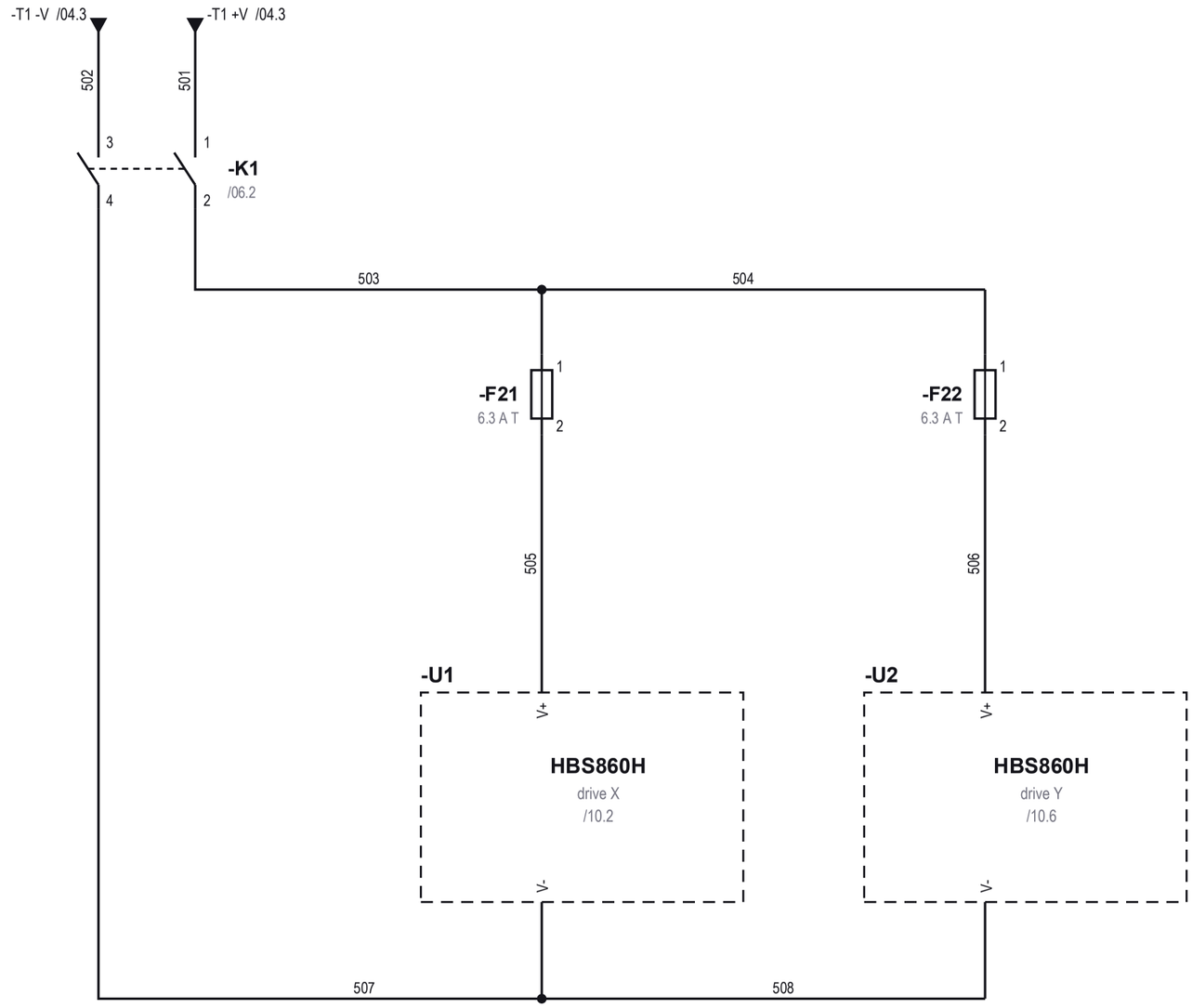
MAINS & POWER SUPPLIES

ไฟ AC และแหล่งจ่ายไฟ

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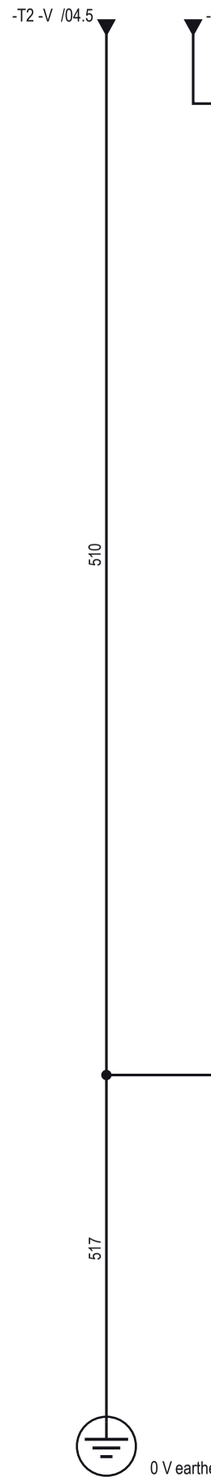
SHEET 04 of 22

MOTOR SUPPLY 60 V DC (switched by -K1)



Note: -U1/-U2 V- also connected via twin ferrule at -K1:4 (508).

CONTROL SU



60 V NOTES

Drive supply lines: twist + and - pairs, route in power duct.
Drive input capacitors: after -K1 opens wait 3 s before re-closing (software 'KM1 reset policy').

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AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

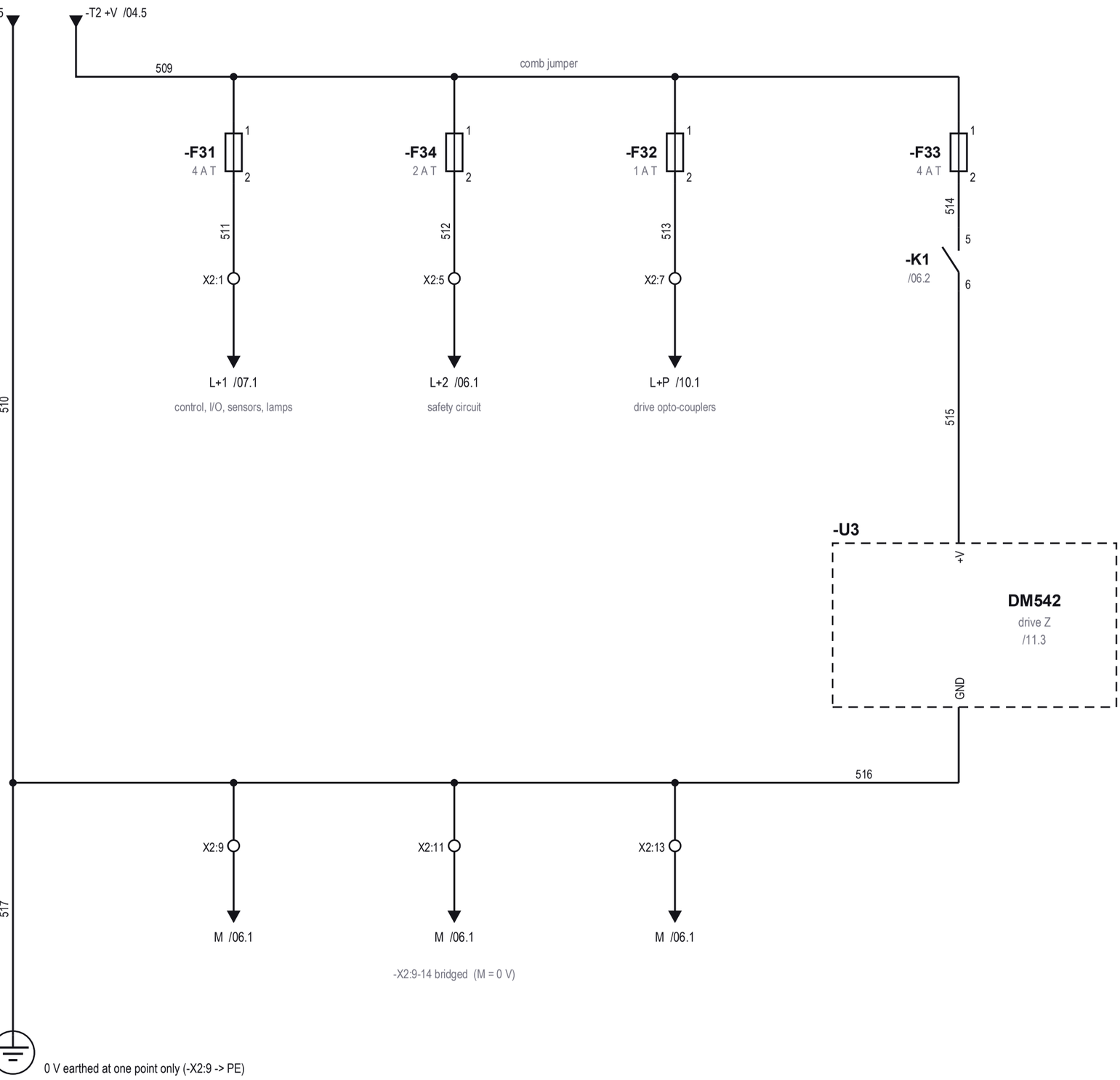
Electrical schematic | IEC 61082 / 81346

SHEET TITLE

DC POWER DI

จ่าย

ONTROL SUPPLY 24 V DC (PELV)



NOTES

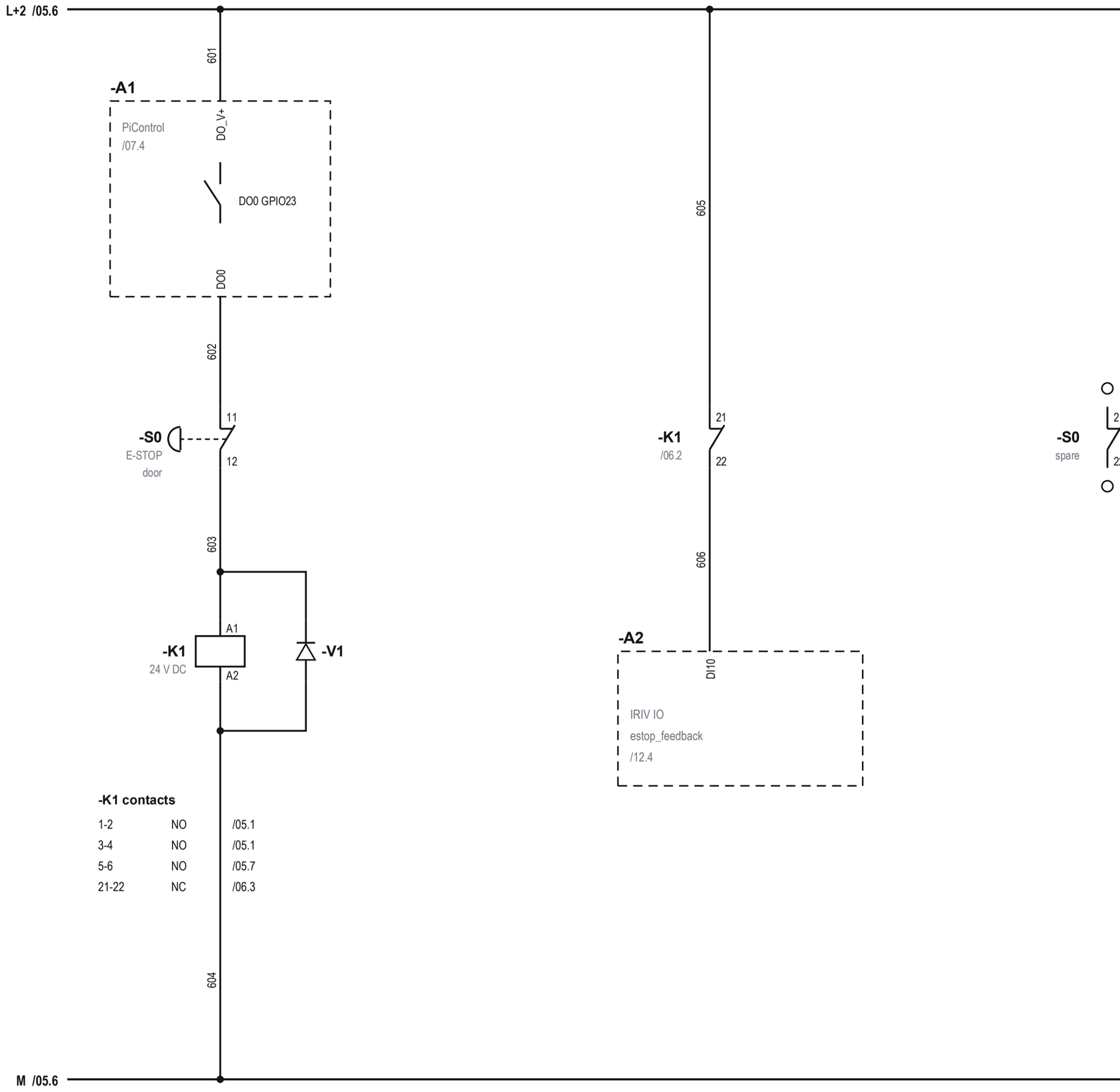
-K1 is a 4-pole DC-rated contactor: poles 1-2/3-4 switch 60 V, pole 5-6 switches drive Z.
Fuses -F21/-F22 protect the drive wiring; -T1 limits total current to 6.7 A.
L+1, L+2, L+P are separately fused so a short in the field does not trip the safety circuit.

WER DISTRIBUTION 60 V / 24 V

จ่ายไฟ DC 60 V และ 24 V

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Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

SAFETY C

วงจร

SAFETY FUNCTION

1 -S0 opens the -K1 coil directly (hard-wired, stop cat. 0, EN 60204-1 9.2.2). No software in this path.
2 -A1 DO0 = software enable. Released by the safety daemon on: drive alarm, Modbus timeout > 350 ms, E-stop request.
3 -K1 removes +60 V / 0V60 from -U1, -U2 and +24 V from -U3.
4 -K1 21-22 -> -A2 DI10: 1 = contactor released (safe). Discrepancy DO0=1 and DI10=1 for > 200 ms = K1 fault.
5 Restart: release -S0, clear alarms, software sets DO0 after 3 s wait (drive capacitor discharge).
6 -V1 suppresses coil switch-off voltage.

Recommendation: for PL d (ISO 13849-1) replace the direct coil circuit with a monitored safety relay using -S0 11-12 and 21-22 (two channels) and -K1 feedback.



-S0
spare

reserve for safety relay

สรุปภาษาไทย

เมื่อกดปุ่ม E-Stop คอยล์ -K1 ถูกตัดทันทีโดยไม่ผ่านซอฟต์แวร์
ไฟ 60 V ของไดรเวอร์ X/Y และ 24 V ของไดรเวอร์ Z ถูกตัด
คอนโทรลเลอร์ยังมีไฟ เพื่อแสดงสถานะและบันทึกเหตุการณ์

FETY CIRCUIT E-STOP / -K1

วงจรความปลอดภัย E-STOP

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L+1 /05.5

703

704

707

-U1

ALM+

PEND+

HBS860H

/10.2

-U2

ALM+

HBS860H

/10.6

ALM-

PEND-

ALM-

705

706

709

-A1

GPIO13

GPIO27

GPIO17

DI0

DI2

DI1

-T3

PSU 5.1 V

USB-C 3 A

/04.7

USB-C

-W701

USB-C 5V

USB-C 3 A

Cytron IRIV PiControl CM4

Raspberry Pi CM4 4 GB / 32 GB eMMC

isolated 24 V I/O

LAN / USB /08.4

DI_COM

711

M /05.6

PAPON

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AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

MAIN CONTROLLE

708

PEND+

HBS860H

/10.6

PEND-

710

O17

GPIO22

DI3

701

24V

DO_V+ → L+2 /05.6

DO0 GPIO23 → -S0 / -K1 /06.2

DO1 GPIO24 spare

DO2 GPIO25 spare

DO3 GPIO16 spare

0V

702

NOTES

ALM / PEND are opto outputs
of the drives, sourced from
L+1 into the isolated inputs.
ALM wired fail-safe: open
circuit = alarm.
Verify output logic of
HBS860H (engineering notes).

ontrol CM4

3 / 32 GB eMMC
/ I/O
/08.4

TROLLER -A1 POWER & LOCAL I/O

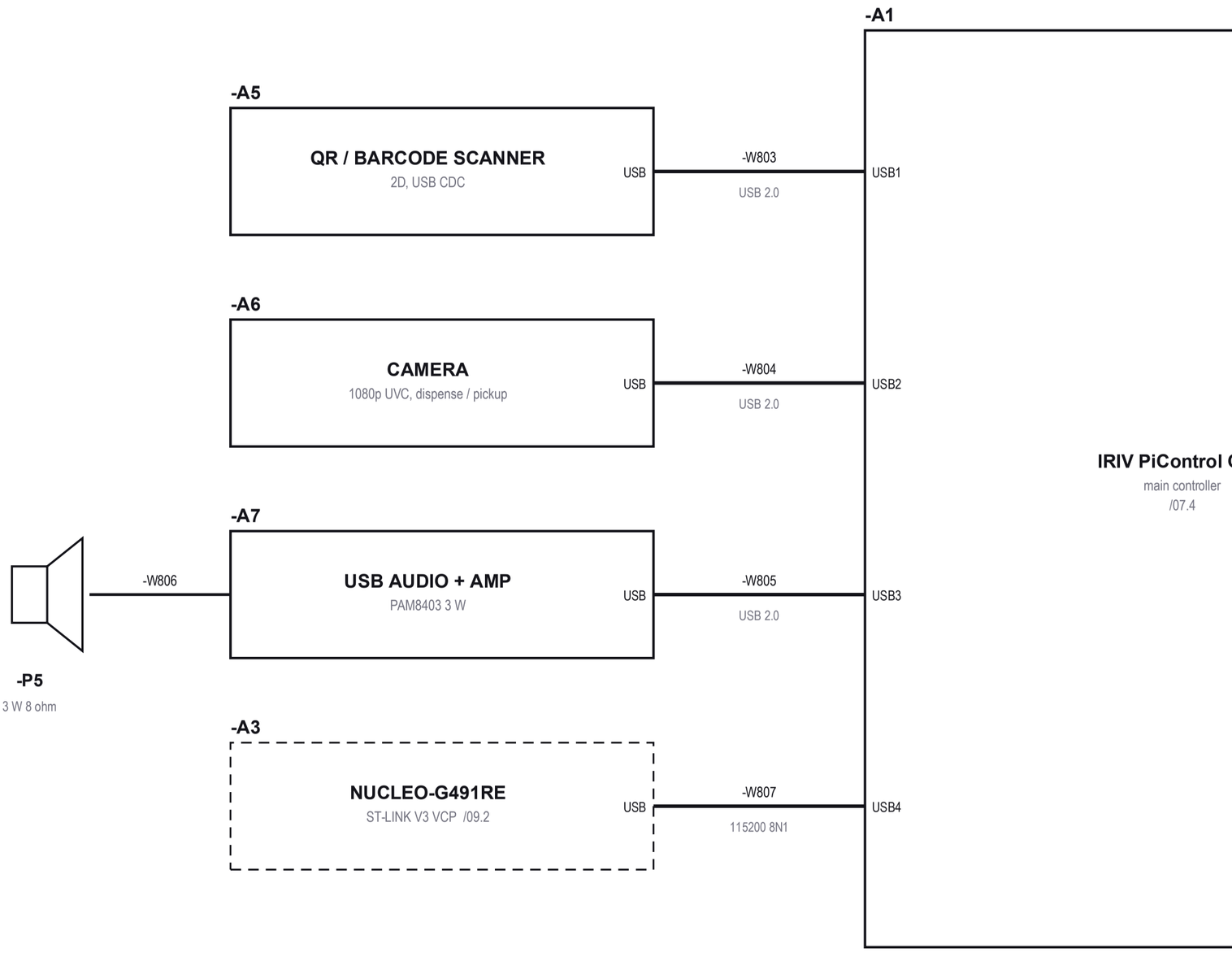
คอนโทรลเลอร์หลัก

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NETWORK & ADDRESSES

Port	Address	Function
-A1 eth0	192.168.70.80	MGMT LAN - HMI kiosk -A8 (192.168.70.81), Flask REST port 80
-A1 eth1	10.0.0.2	OT LAN - Modbus TCP master
-A2	10.0.0.10 : 502	Modbus TCP server, unit ID 255, polled every 20 ms
-A1 USB4 <-> -A3	115200 8N1	Safe-Link v3 over ST-LINK VCP, CRC-16
-A1 (cloud)	broker.emqx.io:1883	MQTT telemetry



Physical separation: ETH0 (MGMT) and ETH1 (OT) are different subnets; no routing between them.

LAN and USB cables are routed in the signal duct, >= 100 mm from 60 V and AC wiring.

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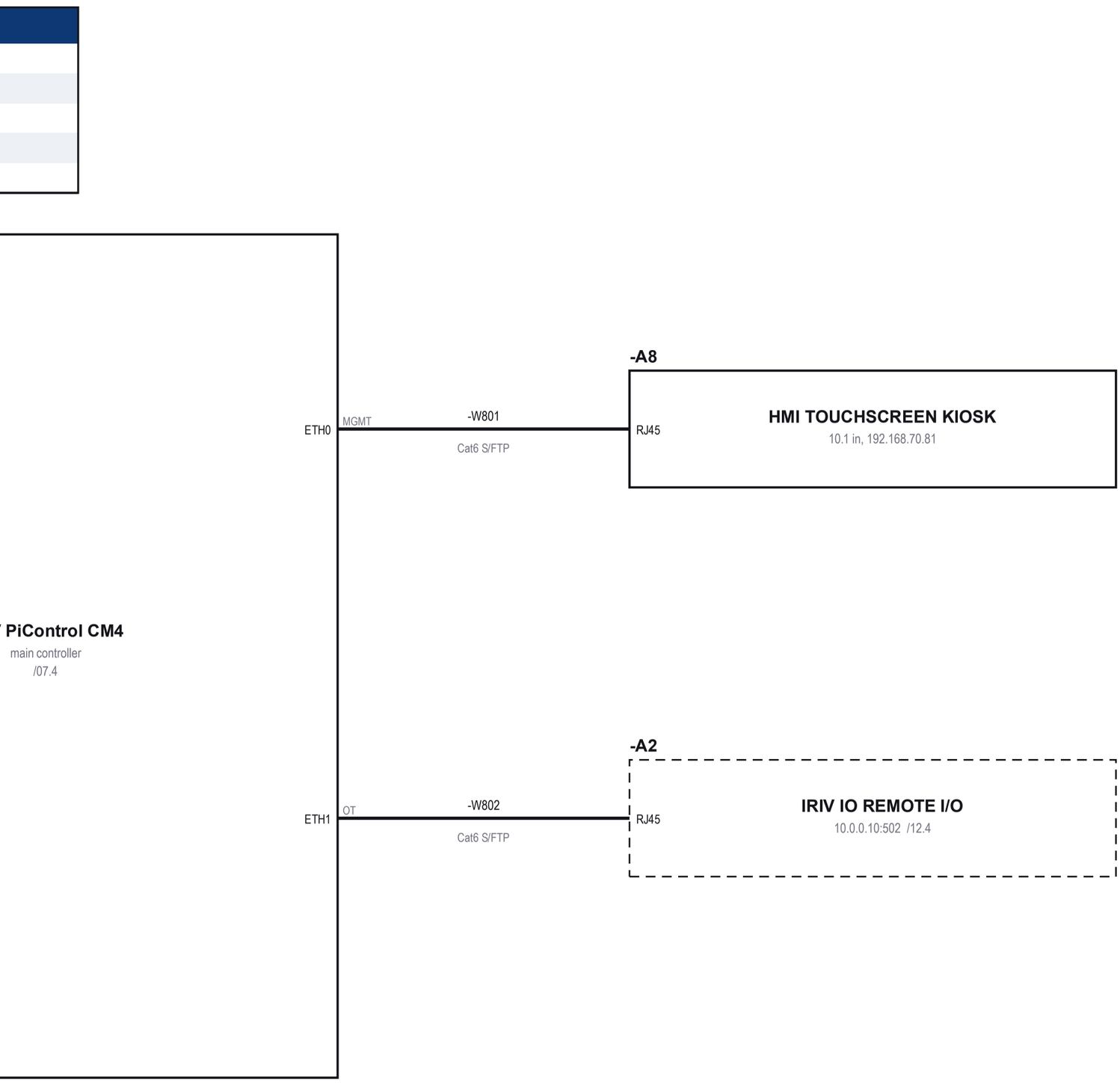
Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

COMMUNICA

การสื่



Service	Description
auto-parts-web-iriv	HMI web UI (Flask REST, port 80)
auto-parts-controller	Core safety interlock / state machine
IPC	Unix socket /run/auto-parts/ctrl.sock
Database	SQLite slots.json / machine_config

MUNICATION & PERIPHERALS

การสื่อสารและอุปกรณ์ต่อพ่วง

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-A3

NUCLEO-G491RE
STM32G491RE 170 MHz
hardware timer pulses
3.3 V logic

/08.2 ← USB ST-LINK
-W807 to -A1

PA8 (TIM1_CH1 STEP X)	CN10-23	901	IN1
PB0 (DIR X)	CN7-34	902	IN2
PA9 (TIM1_CH2 STEP Y)	CN10-21	903	IN3
PB1 (DIR Y)	CN10-24	904	IN4
PA5 (TIM2_CH1 STEP Z)	CN10-11	905	IN5
PB2 (DIR Z)	CN10-22	906	IN6
GND	CN10-20	907	GND

-A4

NMOS SINK BOARD
6 ch AO3400A
open drain 24 V

OUT1 OUT2 OUT3 OUT4 OUT5 OUT6

908 → M /05.6

-W901 jumper leads < 20 cm, GND lead next to signal leads.

FIRMWARE NOTES

Firmware gates: boot DISARMED, 10 Hz - 50 kHz, max. 1 000 000 steps / block, heartbeat timeout 500 ms -> stop, E-stop command stops TIM1/TIM2 immediately, one axis active at a time, S-curve profile, CRC-16-CCITT on every command.
PA5 = user LED LD2 on NUCLEO: LED function disabled in firmware.

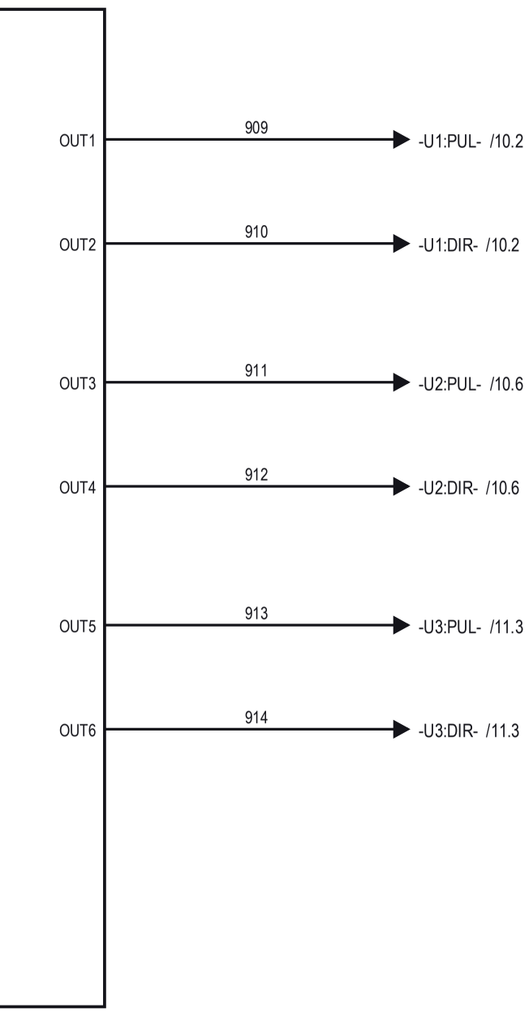
PAPON

DRAFT FOR REVIEW

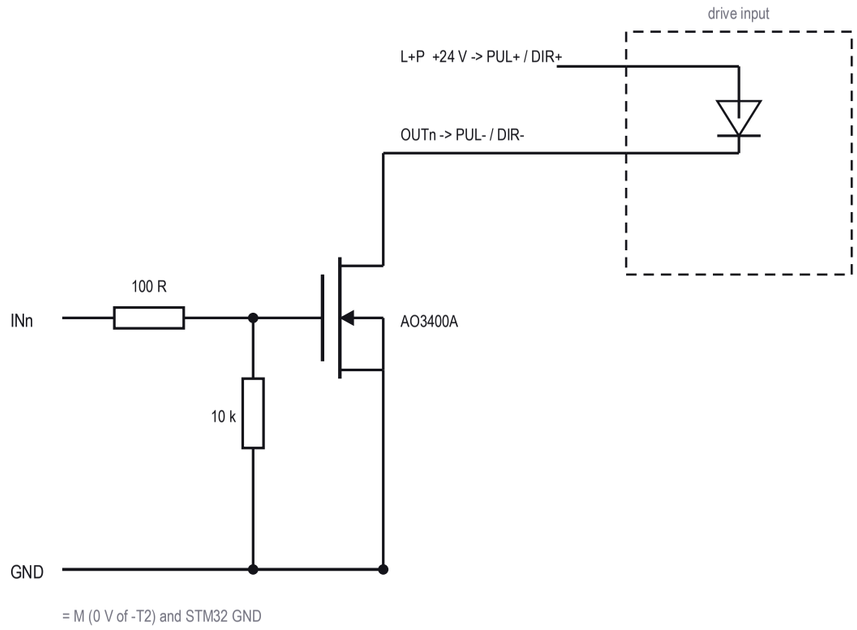
PROJECT
AUTO ELECTRONIC PARTS BOX
Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE
MOTION CO-PROCES
ไมโครคอนโ



TYPICAL CHANNEL (x6)



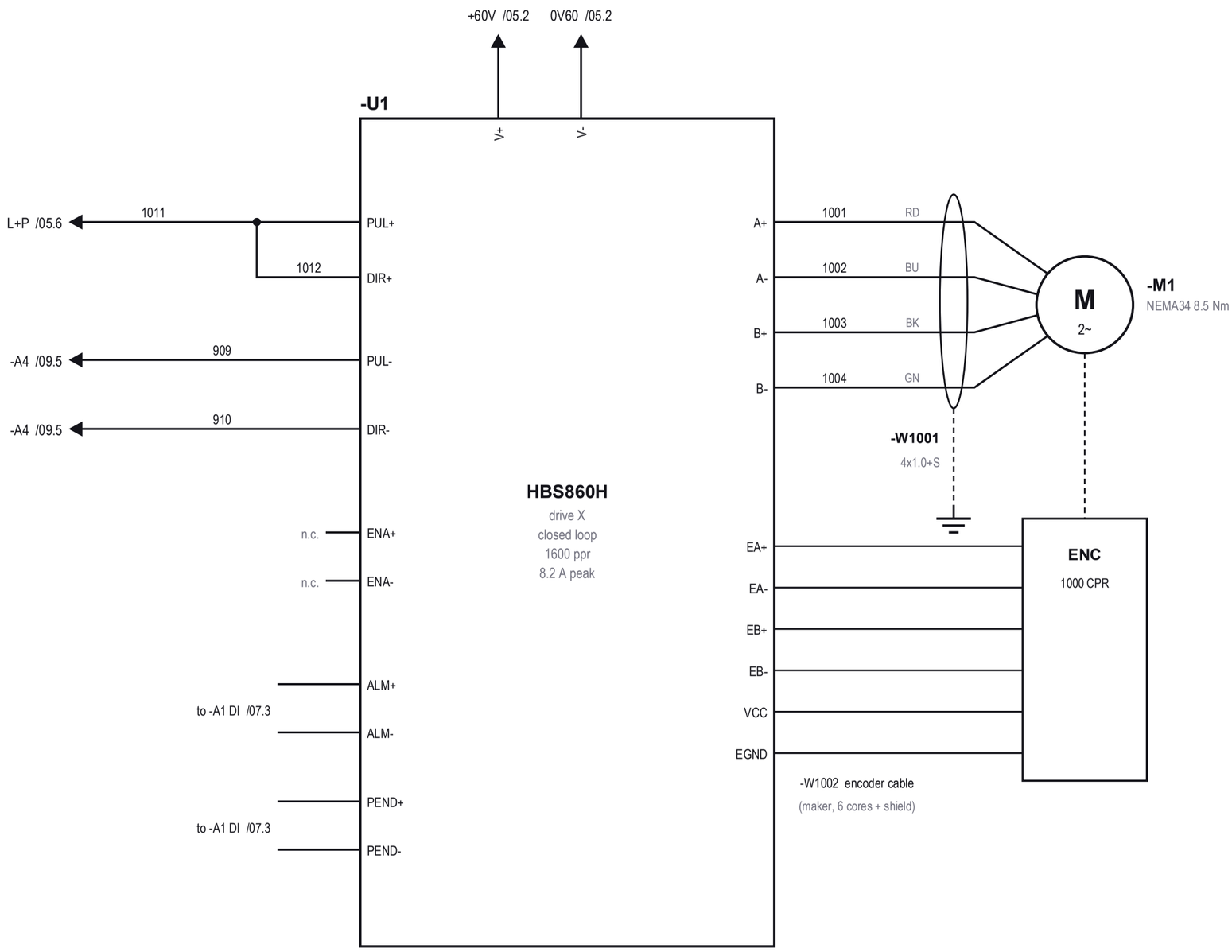
= M (0 V of -T2) and STM32 GND

I_LED = (24 - 1.6 - 0.1) V / R_in approx. 11 mA
(R_in approx. 2 kohm inside drive - verify)
Pulse width >= 2.5 us, DIR setup >= 5 us
Max. step frequency 50 kHz (firmware clamp)

ROCESSOR -A3 & LEVEL SHIFTER -A4

ไมโครคอนโทรลเลอร์และบอร์ดขับสัญญาณ

	DOC. NO. **AUTO-Parts-E01**	DRAWN PAPON	SHEET **09** of 22
	REVISION **4.0**	CHECKED	
	DATE **2026-09-26**	APPROVED	



AXIS DATA

Horizontal carriage, HTD 5M 25 mm belt, pulley pitch dia. 63.66 mm
Kinematics 78.43 steps/mm (1600 ppr), 450 mm/s, 800 mm/s2
DIP: 1600 pulse/rev, peak 8.2 A; tuning via RS232 ProTuner
Motor 86HBS85 closed loop, shaft 14 mm keyed

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

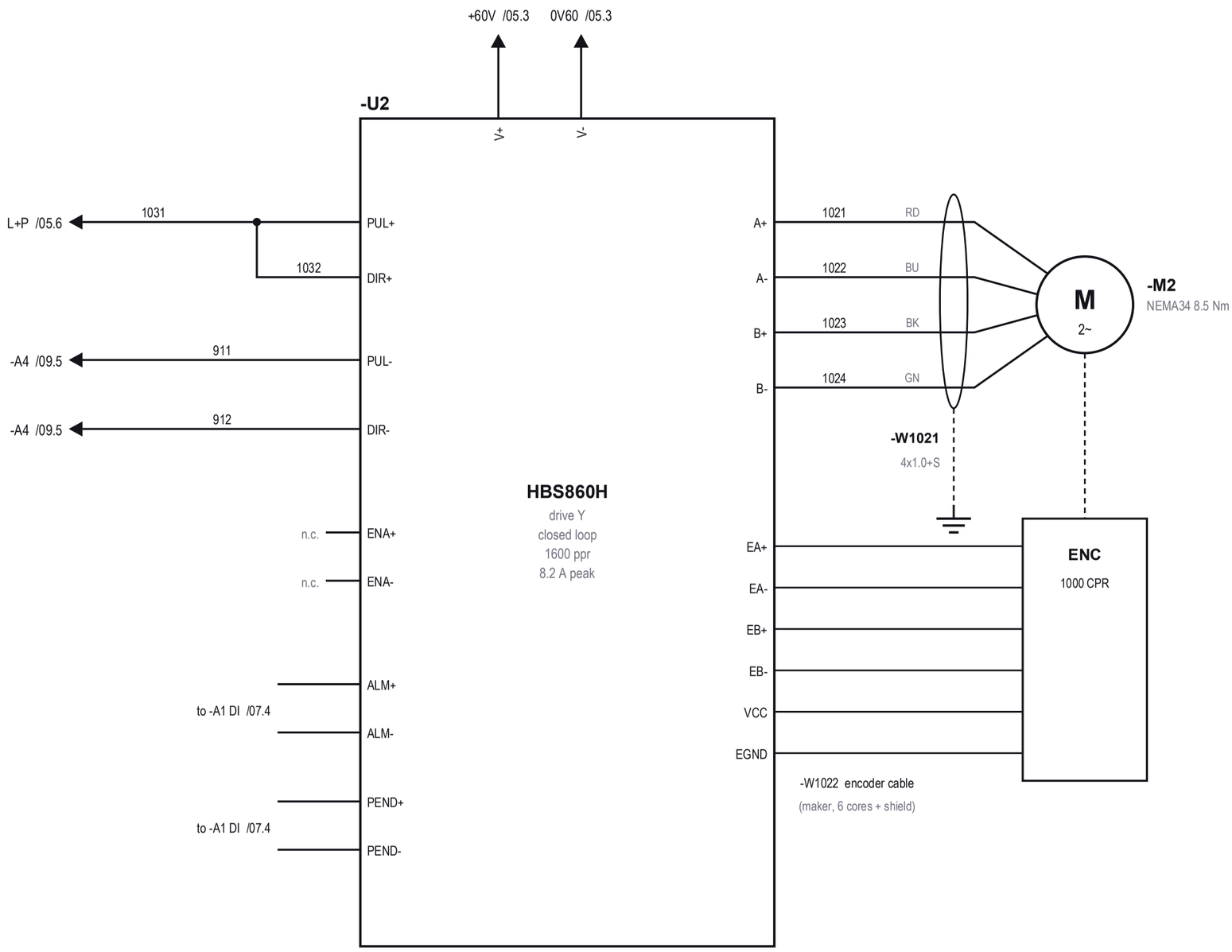
Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

STEPPER D

ไดร



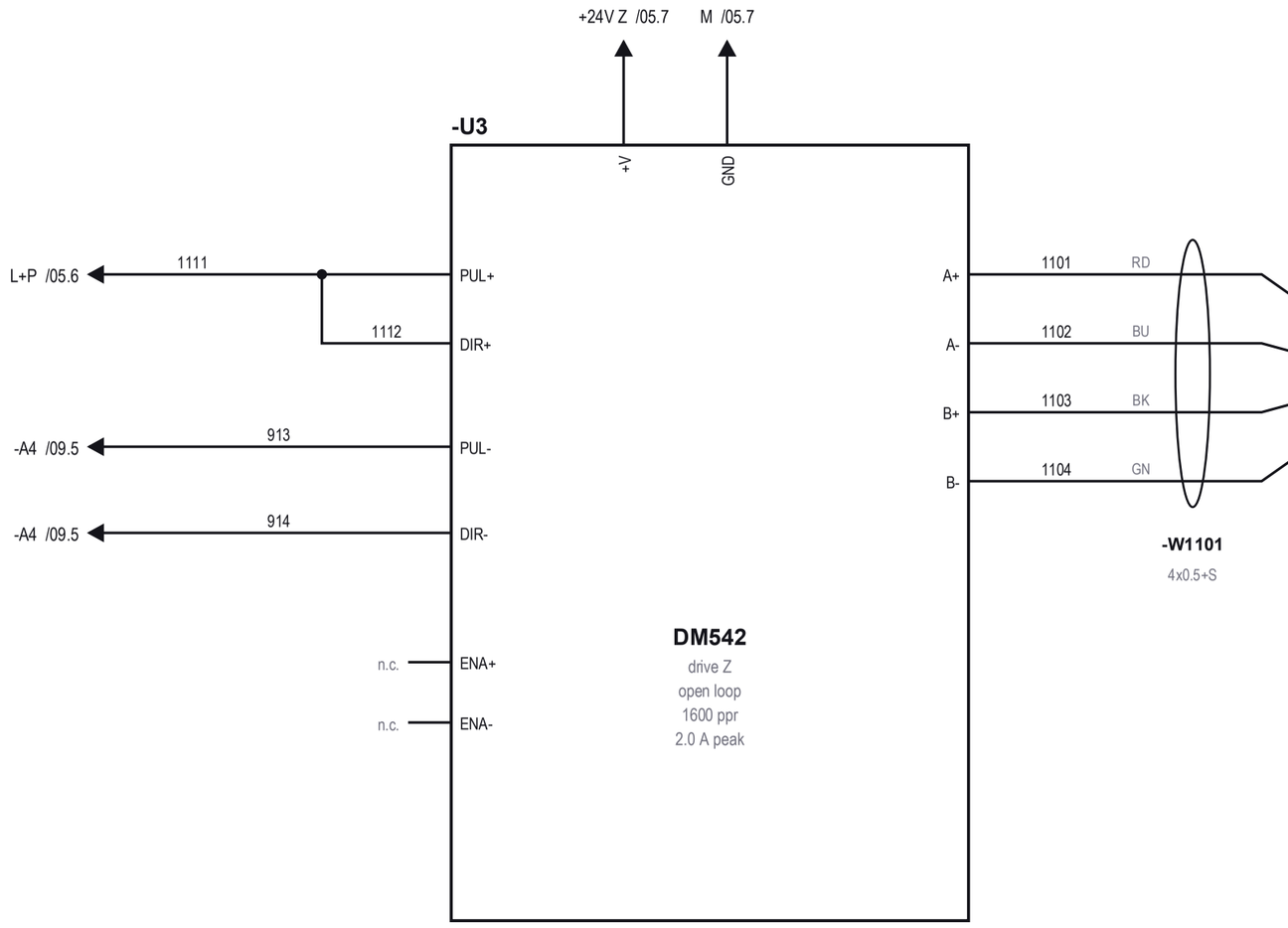
AXIS DATA

Vertical lift elevator, SFU1605 ball screw, lead 5 mm, direct coupling
Kinematics 320.00 steps/mm (1600 ppr), 120 mm/s, 600 mm/s2
DIP: 1600 pulse/rev, peak 8.2 A; brake / holding torque holds elevator
Motor 86HBS85 closed loop, shaft 14 mm keyed

PPER DRIVES X / Y (-U1 / -U2)

ไดรเวอร์มอเตอร์แกน X / Y

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AXIS DATA

V-slot lead screw actuator, T8x8 (pitch 2 mm, 4 starts, lead 8 mm/rev), stroke 150 mm
Kinematics 200.00 steps/mm (1600 ppr), push 80 mm/s, retract 100 mm/s
DIP SW1-SW4 current: peak 2.0 A | DIP SW5-SW8 microstep: 1600 pulse/rev
Supply +24 V Z is switched by -K1 5-6: E-stop now also stops axis Z (new in Rev 4.0).
Limits -B5 (retract) / -B6 (extend) / -B7 (home) on sheet 12.

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

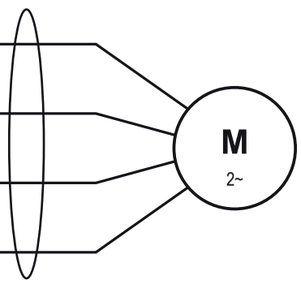
Electrical schematic | IEC 61082 / 81346

SHEET TITLE

STEPPER DRI

ไดรเวอร์

	5	6	7	8



-M3
NEMA17 0.45 Nm
lead screw T8x8

-W1101
4x0.5+S

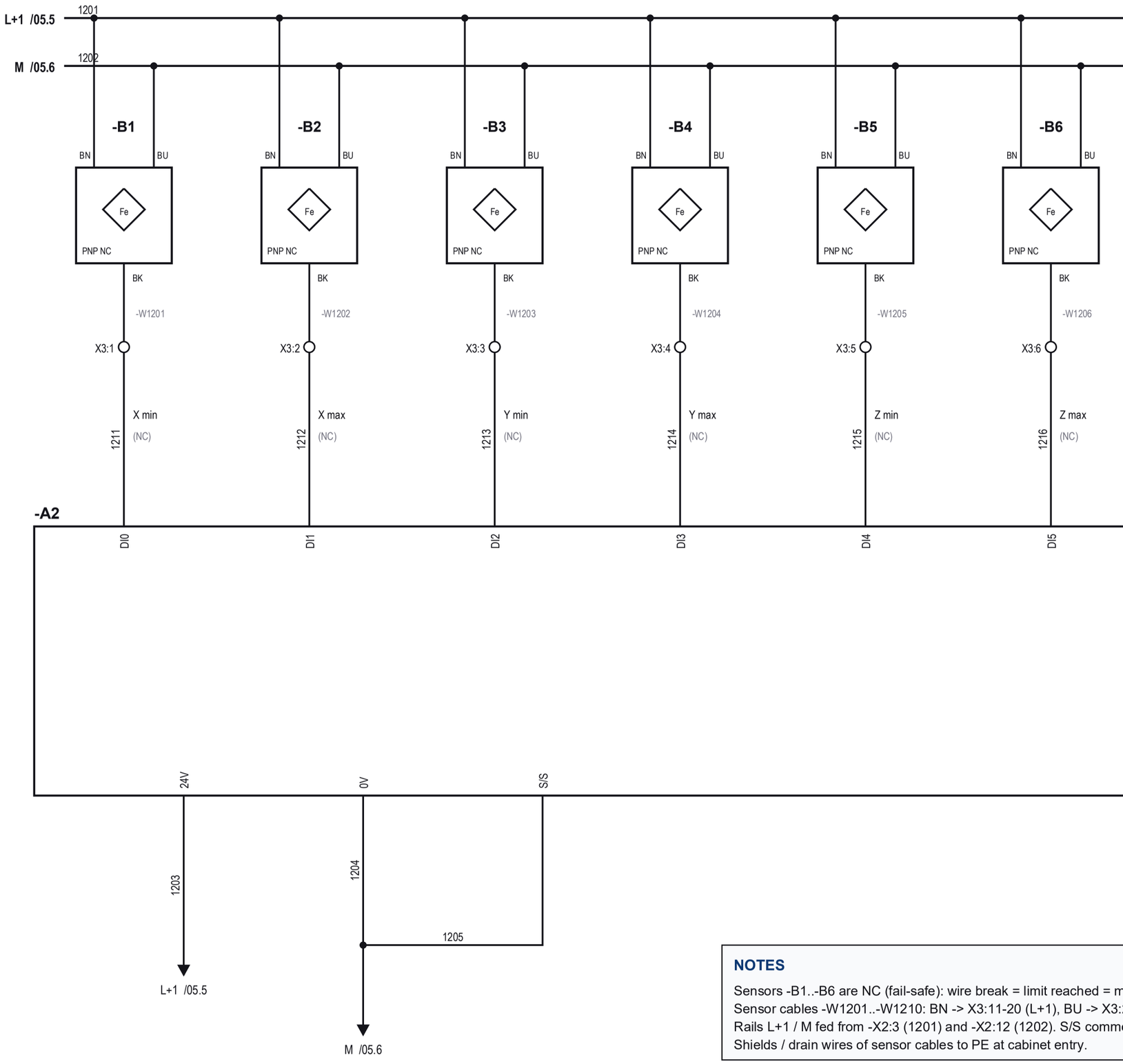
mm

Rev 4.0).

ER DRIVE Z (-U3) DISPENSER

ไดรเวอร์มอเตอร์แกน Z (ดันสินค้า)

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NOTES

Sensors -B1..-B6 are NC (fail-safe): wire break = limit reached = m
Sensor cables -W1201..-W1210: BN -> X3:11-20 (L+1), BU -> X3:
Rails L+1 / M fed from -X2:3 (1201) and -X2:12 (1202). S/S comm
Shields / drain wires of sensor cables to PE at cabinet entry.

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

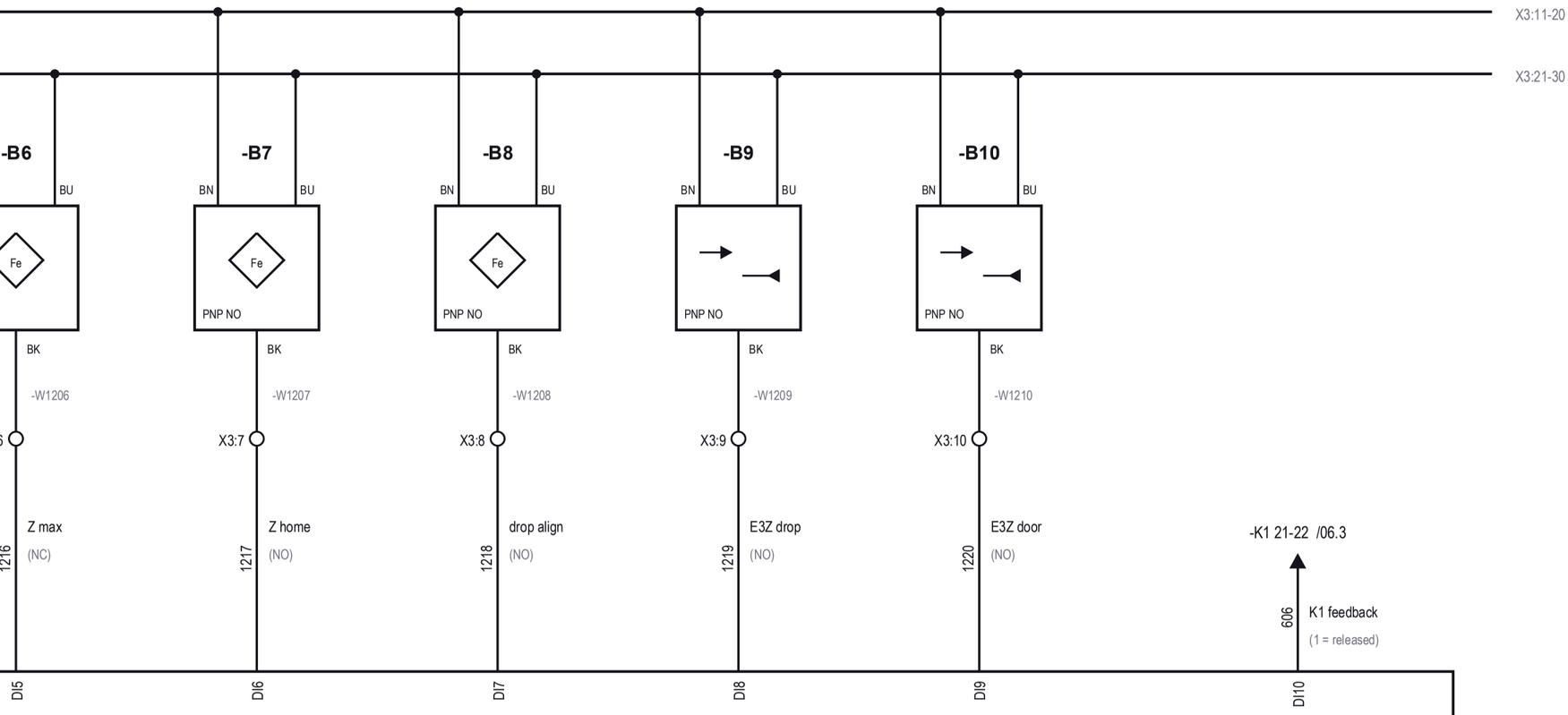
Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

REMOTE I/C

อิน



Cytron IRIV IO - Modbus TCP remote I/O

11 DI opto-isolated 24 V DC, threshold ON > 15 V / OFF < 5 V

/12.4 LAN | outputs /13.3

it reached = motion stops.
+1), BU -> X3:21-30 (M), BK -> X3:1-10 (signal).
02). S/S common tied to 0 V for PNP sensors.
et entry.

OTE I/O -A2 DIGITAL INPUTS อินพุตดิจิทัล (เซนเซอร์)	DOC. NO. **AUTO-Parts-E01**	DRAWN PAPON	SHEET **12** of 22
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1	2	3	4

L+1 /05.5

1301

-A2

DO_COM

IRIV IO /12.4

relay outputs 2 A / 30 V DC

DO0

DO1

-W1301 door cable 7x0.75

(P1..P4 on front door)

1302

1303

X1

-P1

green

X2

X1

-P2

yellow

X2

-P3

red

1307

1308

M /05.6

READY

green DO0

MOVING

yellow DO1

A

PAPON

DRAFT FOR REVIEW

PROJECT

AUTO ELECTRONIC PARTS BOX

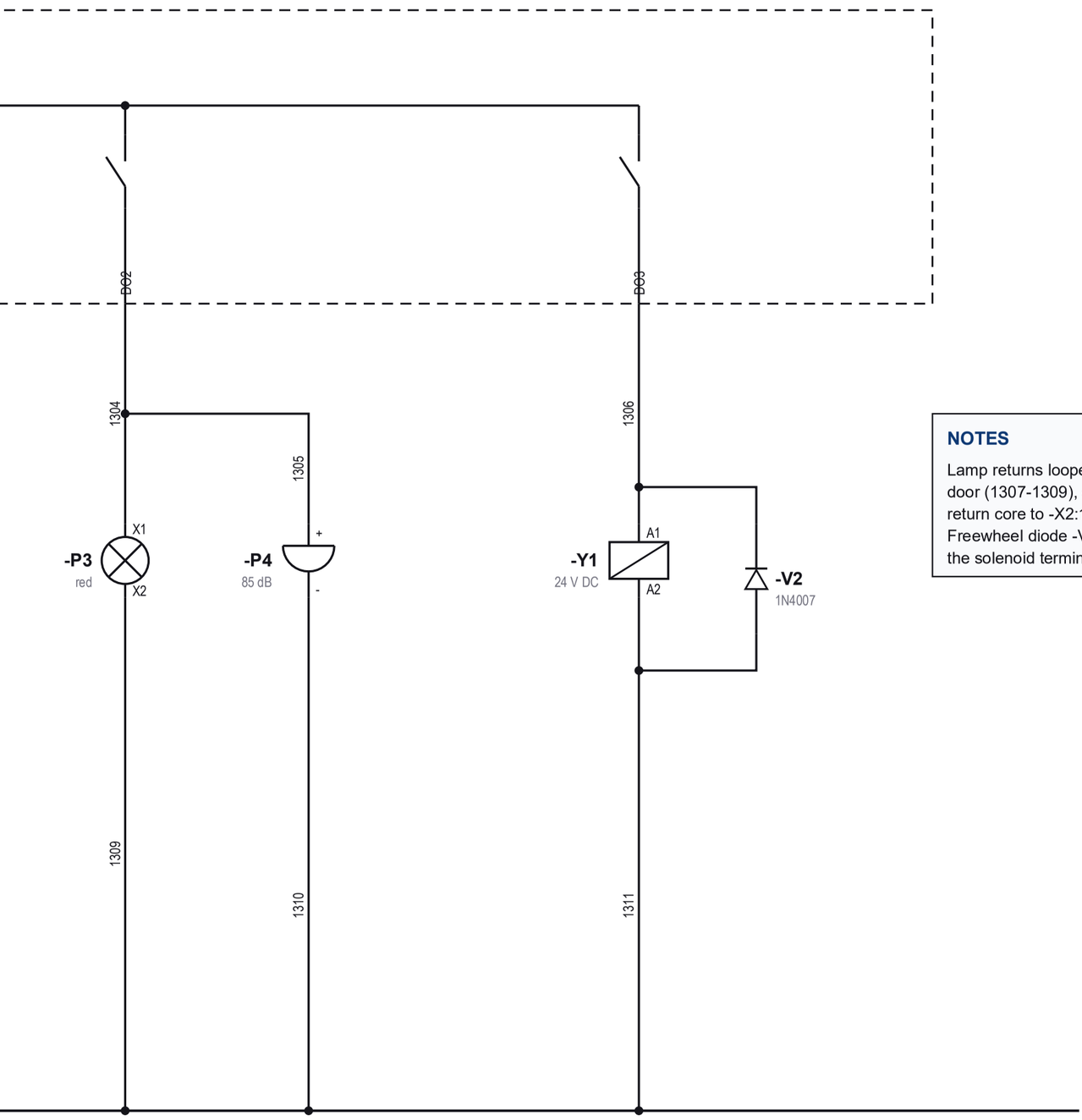
Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

REMOTE I/O

เอาต์พุตดิจิทั



NOTES

Lamp returns looped on door (1307-1309), one return core to -X2:13. Freewheel diode -V2 at the solenoid terminals.

ALARM
red DO2

DISPENSE GATE
DO3

TE I/O -A2 DIGITAL OUTPUTS

อาด์พุตดิจิทัล (ไฟแสดงสถานะ / โซลินอยด์)

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1	2	3	4

-X1 AC distribution 2.5 mm2

Term.	Pot.	Br.	Connections (wire -> device)
1	L'	B1	410 -> -Z1:L', 412 -> -F11:1
2	L'	B1	416 -> -F12:1
3	L'	B1	420 -> -F13:1
4	N'	B2	411 -> -Z1:N', 414 -> -T1:N
5	N'	B2	418 -> -T2:N
6	N'	B2	422 -> -T3:N

-X2 24 V DC distribution

Term.	Pot.	Br.	Connections (wire -> device)
1	L+1	B1	511 -> -F31:2, 701 -> -A1:24V
2	L+1	B1	703 -> -U1:ALM+, 707 -> -U2:ALM+
3	L+1	B1	1201 -> -X3:11, 1203 -> -A2:24V
4	L+1	B1	1301 -> -A2:DO_COM
5	L+2	B2	512 -> -F34:2, 601 -> -A1:DO_V+
6	L+2	B2	605 -> -K1:21
7	L+P	B3	513 -> -F32:2, 1011 -> -U1:PUL+
8	L+P	B3	1031 -> -U2:PUL+, 1111 -> -U3:PUL+
9	M	B4	510 -> -T2:-V, 517 -> -PE
10	M	B4	516 -> -U3:GND, 604 -> -K1:A2
11	M	B4	702 -> -A1:0V, 711 -> -A1:DI_COM
12	M	B4	908 -> -A4:GND, 1202 -> -X3:21
13	M	B4	1204 -> -A2:0V, 1310 -> -P4:-
14	M	B4	1311 -> -Y1:A2

Br. = bridge group (plug-in jumper). Max. 2 conductors per terminal (twin ferrule).

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

TERMINAL

-X3 field sensors

Term.	Pot.	Br.	Connections
1	sig		1211 -> -A2:DI0, -W1201 BK -> -B1
2	sig		1212 -> -A2:DI1, -W1202 BK -> -B2
3	sig		1213 -> -A2:DI2, -W1203 BK -> -B3
4	sig		1214 -> -A2:DI3, -W1204 BK -> -B4
5	sig		1215 -> -A2:DI4, -W1205 BK -> -B5
6	sig		1216 -> -A2:DI5, -W1206 BK -> -B6
7	sig		1217 -> -A2:DI6, -W1207 BK -> -B7
8	sig		1218 -> -A2:DI7, -W1208 BK -> -B8
9	sig		1219 -> -A2:DI8, -W1209 BK -> -B9
10	sig		1220 -> -A2:DI9, -W1210 BK -> -B10
11	L+1	B1	1201 -> -X2:3, -W1201 BN -> -B1
12	L+1	B1	-W1202 BN -> -B2
13	L+1	B1	-W1203 BN -> -B3
14	L+1	B1	-W1204 BN -> -B4
15	L+1	B1	-W1205 BN -> -B5
16	L+1	B1	-W1206 BN -> -B6
17	L+1	B1	-W1207 BN -> -B7
18	L+1	B1	-W1208 BN -> -B8
19	L+1	B1	-W1209 BN -> -B9
20	L+1	B1	-W1210 BN -> -B10
21	M	B2	1202 -> -X2:12, -W1201 BU -> -B1
22	M	B2	-W1202 BU -> -B2
23	M	B2	-W1203 BU -> -B3
24	M	B2	-W1204 BU -> -B4
25	M	B2	-W1205 BU -> -B5
26	M	B2	-W1206 BU -> -B6
27	M	B2	-W1207 BU -> -B7
28	M	B2	-W1208 BU -> -B8
29	M	B2	-W1209 BU -> -B9
30	M	B2	-W1210 BU -> -B10

1	2	3	4

CHANGES VERSUS REV 3.0 / สิ่งที่แก้ไขจาก Rev 3.0

1. One document number, one wire-numbering system and IEC 81346-2 device tags on all sheets.
   ใช้เลขที่เอกสาร ระบบเลขสาย และรหัสอุปกรณ์ตาม IEC 81346-2 ชุดเดียวกันทุกหน้า
2. Surge arrester -F1 drawn and wired in parallel (L-PE, N-PE) after -Q1, not in series.
   อุปกรณ์กันฟ้าผ่า -F1 ต่อแบบขนานหลัง -Q1 (เดิมวาดเป็นอนุกรม)
3. Branch fuses added: -F11..-F13 per power supply, -F21/-F22 per drive, -F31..-F34 per 24 V circuit.
   เพิ่มฟิวส์แยกวงจร: ต่อ PSU, ต่อไดรเวอร์ และต่อวงจร 24 V
4. -K1 is now a 4-pole DC-rated contactor. Pole 5-6 also removes 24 V from drive Z (-U3); in Rev 3.0 axis Z stayed live after E-stop.
   -K1 เปลี่ยนเป็นคอนแทคเตอร์ 4 ขั้วพิกัด DC และตัดไฟไดรเวอร์แกน Z ด้วย (เดิมกด E-Stop แล้วแกน Z ยังมีไฟ)
5. E-stop -S0 hard-wired in series with -A1 DO0 in the -K1 coil circuit (stop category 0, EN 60204-1 9.2.2).
   ปุ่ม E-Stop -S0 ต่ออนุกรมกับคอยล์ -K1 โดยตรง ไม่ผ่านซอฟต์แวร์
6. Coil suppressor -V1 on -K1 and freewheel diode -V2 on solenoid -Y1.
   เพิ่มไดโอดกันแรงดันย้อนที่คอยล์ -K1 และโซลินอยด์ -Y1
7. 0 V (M) of the 24 V supply earthed at one point (-X2:9 to PE) = PELV.
   ต่อ 0 V ของ 24 V ลงดินจุดเดียว (PELV)
8. Terminal strips -X1 (AC), -X2 (24 V) and -X3 (sensors) introduced; max. 2 conductors per terminal.
   เพิ่มเทอร์มินอล -X1, -X2, -X3 และจำกัดไม่เกิน 2 สายต่อขั้ว
9. Wire colours per EN 60204-1: DC control dark blue, 0 V dark blue/white, PE green/yellow.
   สีสายตาม EN 60204-1

OPEN POINTS - TO BE VERIFIED ON SITE / ต้องตรวจสอบหน้างาน

- ☐ Confirm HBS860H / DM542 PUL/DIR inputs accept 24 V directly. If 5 V only, add 2.2 kohm 0.5 W in series with PUL+ and DIR+.
  ตรวจว่าไดรเวอร์รับสัญญาณ 24 V ได้ ถ้าไม่ได้ต้องใส่ R 2.2 kΩ
- ☐ Confirm the ALM/PEND output logic of HBS860H (conducting = normal?) and set software accordingly.
  ยืนยันตรรกะสัญญาณ ALM/PEND ของไดรเวอร์
- ☐ Confirm IRIV PiControl supply: USB-C 5 V (as drawn) or DC terminal; and DO type (sourcing).
  ยืนยันวิธีจ่ายไฟ PiControl และชนิดเอาต์พุต DO
- ☐ Confirm IRIV IO input common (S/S) wiring for PNP sensors and relay output rating (2 A / 30 V DC).
  ยืนยันการต่อขั้วร่วม S/S ของ IRIV IO
- ☐ Select -K1 with a documented DC-1 rating at 60 V DC (switching capacitive drive input).
  เลือก -K1 ที่มีพิกัดตัดไฟ DC 60 V จริง
- ☐ Confirm NC/NO variants of -B1..-B10 and cable lengths on site.
  ตรวจชนิด NC/NO ของเซนเซอร์และความยาวสายหน้างาน
- ☐ Power source of HMI kiosk -A8 is not shown in Rev 3.0 - to be confirmed.
  ยังไม่ทราบแหล่งจ่ายไฟของจอ HMI -A8
- ☐ -S0 contact 21-22 is spare. For PL d (ISO 13849-1) a monitored safety relay is recommended.
  หน้าสัมผัส 21-22 ของ -S0 ว่าง แนะนำใช้ Safety relay ในอนาคต
- ☐ ENA inputs of -U1..-U3 not connected: drives are enabled whenever powered.
  ขา ENA ไม่ได้ต่อ ไดรเวอร์จะทำงานทันทีเมื่อมีไฟ

PAPON **DRAFT FOR REVIEW**	PROJECT **AUTO ELECTRONIC PARTS BOX** Auto Electronic Parts Box Electrical schematic | IEC 61082 / 81346	SHEET TITLE **ENGINEERING** หมายเหตุทาง

	5	6	7	8

STATUS

This drawing set was re-created from the Rev 3.0 AI-generated sheets. Values not stated there were chosen according to common practice and marked 'verify'.

Must be checked and signed by a qualified electrical engineer before it is issued as AS-BUILT.

แบบชุดนี้จัดทำใหม่จากข้อมูล Rev 3.0
ค่าที่ไม่มีในต้นฉบับเลือกตามแนวปฏิบัติทั่วไป
ต้องให้วิศวกรไฟฟ้าตรวจสอบและลงนาม
ก่อนออกเป็นแบบ As-Built

EERING NOTES & OPEN POINTS

หมายเหตุทางวิศวกรรมและจุดที่ต้องตรวจสอบ

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Item	Tag	Qty	Description	Type
1	-X0	1	Power inlet IEC C14 with fuse holder, fuse 10 A T 5x20	IEC C1
2	-Q1	1	Miniature circuit breaker 2-pole C16, 6 kA (main switch)	2P C16
3	-F1	1	Surge protective device Type 2, 1P+N, Uc 275 V, In 10 kA	SPD T2
4	-Z1	1	EMI line filter 1-phase 20 A	CW4L2
5	-X1	1	Terminal strip AC distribution 2.5 mm2 (L x3, N x3, bridged)	Feed-th
6	-F11	1	Fuse terminal 5x20, fuse 6.3 A T (supply -T1)	Fuse te
7	-F12	1	Fuse terminal 5x20, fuse 2 A T (supply -T2)	Fuse te
8	-F13	1	Fuse terminal 5x20, fuse 1 A T (supply -T3)	Fuse te
9	-T1	1	Power supply 60 V DC 6.7 A 400 W (motor supply X/Y)	S-400-6
10	-T2	1	Power supply 24 V DC 5 A 120 W DIN rail (control)	NDR-1
11	-T3	1	Power supply 5.1 V DC 3 A USB-C (PiControl CM4)	RPi 15
12	-K1	1	Safety contactor 4-pole, DC-rated (DC-1 >= 16 A at 60 V), coil 24 V DC, aux 1 NC	4P DC
13	-V1	1	Coil suppressor module (diode) for -K1, 24 V DC	Suppre
14	-S0	1	Emergency stop push button 40 mm, twist release, 2 NC	XB5AS
15	-F21	1	Fuse terminal 5x20, fuse 6.3 A T (drive -U1)	Fuse te
16	-F22	1	Fuse terminal 5x20, fuse 6.3 A T (drive -U2)	Fuse te
17	-F31	1	Fuse terminal 5x20, fuse 4 A T (L+1 control / I/O)	Fuse te
18	-F32	1	Fuse terminal 5x20, fuse 1 A T (L+P pulse supply)	Fuse te
19	-F33	1	Fuse terminal 5x20, fuse 4 A T (drive -U3 supply)	Fuse te
20	-F34	1	Fuse terminal 5x20, fuse 2 A T (L+2 safety circuit)	Fuse te
21	-X2	1	Terminal strip 24 V DC distribution (14 terminals, bridged groups)	Feed-th
22	-X3	1	Terminal strip field sensors (30 terminals)	Feed-th
23	-PE	1	PE busbar with cabinet and door bonding	PE bar
24	-A1	1	Main controller Cytron IRIV PiControl CM4 (4 GB / 32 GB eMMC, Wi-Fi)	IRIV Pi
25	-A2	1	Remote I/O Modbus TCP, 11 DI / 4 DO, 24 V DC	IRIV IO
26	-A3	1	Motion co-processor board STM32G491RE	NUCLE
27	-A4	1	6-channel NMOS low-side driver board (3.3 V in, 24 V open drain)	AO340
28	-A5	1	2D QR / barcode scanner, USB CDC	2D sca
29	-A6	1	USB camera 1080p UVC, wide angle	1080p
30	-A7	1	USB audio adapter + amplifier 3 W	PAM84
31	-A8	1	Touchscreen HMI kiosk 10.1 in, Ethernet	10.1 in
32	-U1	1	Closed-loop stepper drive X axis, 24-75 V DC (8.2 A peak)	HBS86
33	-U2	1	Closed-loop stepper drive Y axis, 24-75 V DC (8.2 A peak)	HBS86
34	-U3	1	Stepper drive Z axis (dispenser), 20-50 V DC (4.2 A peak)	DM542
35	-M1	1	Hybrid closed-loop stepper NEMA 34, 8.5 Nm, encoder 1000 CPR (X)	86HBS
36	-M2	1	Hybrid closed-loop stepper NEMA 34, 8.5 Nm, encoder 1000 CPR (Y)	86HBS
37	-M3	1	Stepper NEMA 17 linear actuator, lead screw T8x8, stroke 150 mm (Z)	NEMA
38	-B1	1	Inductive sensor M12 PNP NC, 10-30 V DC - X min (home)	M12 PI
39	-B2	1	Inductive sensor M12 PNP NC - X max	M12 PI
40	-B3	1	Inductive sensor M12 PNP NC - Y min (drop level)	M12 PI
41	-B4	1	Inductive sensor M12 PNP NC - Y max	M12 PI
42	-B5	1	Inductive sensor M12 PNP NC - Z min (retract)	M12 PI
43	-B6	1	Inductive sensor M12 PNP NC - Z max (extend)	M12 PI
44	-B7	1	Inductive sensor M12 PNP NO - Z home reference	M12 PI
45	-B8	1	Inductive sensor M12 PNP NO - product parking / drop align	M12 PI
46	-B9	1	Photoelectric sensor diffuse PNP - product drop verify	E3Z-D8

PAPON

DRAFT FOR REVIEW

PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

PARTS

Type	Maker	Location	Remark
IEC C14 + fuse	Schurter / eq.	Cabinet wall	
2P C16	Schneider / ABB / eq.	Panel	Lockable handle recommended
SPD T2 1P+N	Phoenix / eq.	Panel	
CW4L2-20A-S	Canny Well	Panel	
Feed-through 2.5 mm2	Phoenix / Wago	Panel	
Fuse terminal	Phoenix / eq.	Panel	
Fuse terminal	Phoenix / eq.	Panel	
Fuse terminal	Phoenix / eq.	Panel	
S-400-60	Generic	Panel	
NDR-120-24	Mean Well	Panel	
RPi 15 W USB-C	Raspberry Pi	Panel	DIN-rail 5 V PSU + USB-C lead recommended
4P DC contactor	Schneider / eq.	Panel	Verify DC switching rating
Suppressor	match -K1	Panel	
XB5AS8442 / eq.	Schneider / eq.	Front door	
Fuse terminal	Phoenix / eq.	Panel	
Fuse terminal	Phoenix / eq.	Panel	
Fuse terminal	Phoenix / eq.	Panel	
Fuse terminal	Phoenix / eq.	Panel	
Fuse terminal	Phoenix / eq.	Panel	
Fuse terminal	Phoenix / eq.	Panel	
Feed-through 2.5 mm2	Phoenix / Wago	Panel	
Feed-through 1.5 mm2	Phoenix / Wago	Panel	
PE bar	-	Panel	
IRIV PiControl CM4	Cytron	Panel	
IRIV IO Controller	Cytron	Panel	
NUCLEO-G491RE	STMicroelectronics	Panel	
AO3400A x6	Custom	Panel	
2D scanner	Generic	Front door	
1080p UVC	Generic	Pickup chute	
PAM8403	Generic	Panel	
10.1 in kiosk	Generic	Front door	Power source to be confirmed
HBS860H	Leadshine	Panel	
HBS860H	Leadshine	Panel	
DM542	Leadshine	Panel	
86HBS85	Leadshine	X carriage	
86HBS85	Leadshine	Y elevator	
NEMA 17 + T8x8	Generic	Dispenser	
M12 PNP NC	Omron / eq.	Field	
M12 PNP NC	Omron / eq.	Field	
M12 PNP NC	Omron / eq.	Field	
M12 PNP NC	Omron / eq.	Field	
M12 PNP NC	Omron / eq.	Field	
M12 PNP NC	Omron / eq.	Field	
M12 PNP NO	Omron / eq.	Field	
M12 PNP NO	Omron / eq.	Field	
E3Z-D81	Omron	Field	

PARTS LIST (BOM) (1/2)

รายการอุปกรณ์

DOC. NO. **AUTO-Parts-E01**	DRAWN PAPON	SHEET **16** of 22
REVISION **4.0**	CHECKED	
DATE **2026-09-26**	APPROVED	

1	2	3	4	

Item	Tag	Qty	Description	Type
47	-B10	1	Photoelectric sensor diffuse PNP - pickup door	E3Z-D8
48	-P1	1	Pilot light 22 mm LED 24 V DC green - READY	22 mm
49	-P2	1	Pilot light 22 mm LED 24 V DC yellow - MOVING	22 mm
50	-P3	1	Pilot light 22 mm LED 24 V DC red - ALARM	22 mm
51	-P4	1	Buzzer 24 V DC 85 dB - ALARM	22 mm
52	-P5	1	Loudspeaker 3 W 8 ohm - voice guidance	3 W 8 o
53	-Y1	1	Solenoid 24 V DC - dispense gate	24 V so
54	-V2	1	Freewheel diode 1N4007 across -Y1	1N4007

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

PARTS

Type	Maker	Location	Remark
E3Z-D81	Omron	Field	
22 mm LED	Schneider / eq.	Front door	
22 mm LED	Schneider / eq.	Front door	
22 mm LED	Schneider / eq.	Front door	
22 mm buzzer	Generic	Front door	
3 W 8 ohm	Generic	Front door	
24 V solenoid	Generic	Dispenser	
1N4007	-	At -Y1	

PARTS LIST (BOM) (2/2)

รายการอุปกรณ์

DOC. NO. **AUTO-Parts-E01**	DRAWN PAPON	SHEET **17** of 22	
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1	2	3	4

Cable	From	To	Type	Cores	Length
-W401	Mains socket	-X0	H05VV-F + C13 plug	3G1.5	2.0
-W601	Panel	-S0	LiYY	4x0.75	1.5
-W701	-T3	-A1:USB-C	USB-C power lead 3 A	USB-C	0.5
-W801	-A1:ETH0	-A8	Cat6 S/FTP patch	RJ45	2.0
-W802	-A1:ETH1	-A2	Cat6 S/FTP patch	RJ45	0.5
-W803	-A1:USB1	-A5	USB 2.0	USB-A	2.0
-W804	-A1:USB2	-A6	USB 2.0	USB-A	2.0
-W805	-A1:USB3	-A7	USB 2.0	USB-A	0.5
-W806	-A7	-P5	Speaker cable	2x0.5	1.5
-W807	-A1:USB4	-A3:ST-LINK	USB 2.0	USB	0.5
-W901	-A3	-A4	Jumper leads (Dupont)	7x0.25	0.2
-W1001	-U1	-M1	Motor cable shielded	4x1.0+S	3.0
-W1002	-U1	-M1 encoder	Encoder cable (maker)	6x0.2+S	3.0
-W1021	-U2	-M2	Motor cable shielded	4x1.0+S	3.0
-W1022	-U2	-M2 encoder	Encoder cable (maker)	6x0.2+S	3.0
-W1101	-U3	-M3	Motor cable shielded	4x0.5+S	2.0
-W1301	Panel	-P1..-P4	LiYY	7x0.75	1.5
-W1302	Panel	-Y1	LiYY	2x0.75	2.0
-W1201	-B1	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1202	-B2	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1203	-B3	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1204	-B4	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1205	-B5	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1206	-B6	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1207	-B7	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1208	-B8	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1209	-B9	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0
-W1210	-B10	-X3	Sensor cable PUR (maker)	3x0.34	2.0 / 5.0

PAPON

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AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

C

Length m	Remark
2.0	Mains cord
1.5	E-stop door cable (2 spare)
0.5	
2.0	MGMT LAN 192.168.70.x
0.5	OT LAN 10.0.0.x, Modbus TCP
2.0	QR scanner
2.0	Camera
0.5	Audio
1.5	
0.5	Safe-Link VCP 115200 8N1
0.2	Keep < 20 cm
3.0	Shield 360 deg at PE
3.0	Leadshine extension
3.0	Shield 360 deg at PE
3.0	Leadshine extension
2.0	
1.5	Door cable lamps/buzzer
2.0	Dispense solenoid
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal
2.0 / 5.0	BN=L+, BU=M, BK=signal

CABLE LIST

รายการสายเคเบิล

DOC. NO. **AUTO-Parts-E01**	DRAWN PAPON	SHEET **18**
REVISION **4.0**	CHECKED	of 22
DATE **2026-09-26**	APPROVED	

1	2	3	4

Ctrl	Address	Dir	Device	Signal	Logic	Description
-A1	DI0 (GPIO13)	IN	-U1	X_DRIVE_ALM	NC fail-safe	Drive X alarm
-A1	DI1 (GPIO17)	IN	-U2	Y_DRIVE_ALM	NC fail-safe	Drive Y alarm
-A1	DI2 (GPIO27)	IN	-U1	X_PEND	1 = in position	Drive X in-position
-A1	DI3 (GPIO22)	IN	-U2	Y_PEND	1 = in position	Drive Y in-position
-A1	DO0 (GPIO23)	OUT	-K1	XY_DRIVE_POWER_KM1	1 = enable	Safety contactor enable
-A1	DO1 (GPIO24)	OUT	-	spare		Auxiliary output 1
-A1	DO2 (GPIO25)	OUT	-	spare		Auxiliary output 2
-A1	DO3 (GPIO16)	OUT	-	spare		Auxiliary output 3
-A2	DI0	IN	-B1	x_head_limit	NC fail-safe	X min limit / home
-A2	DI1	IN	-B2	x_tail_limit	NC fail-safe	X max limit
-A2	DI2	IN	-B3	y_head_limit	NC fail-safe	Y min limit (drop level)
-A2	DI3	IN	-B4	y_tail_limit	NC fail-safe	Y max limit
-A2	DI4	IN	-B5	z_head_limit	NC fail-safe	Z min limit (retract)
-A2	DI5	IN	-B6	z_tail_limit	NC fail-safe	Z max limit (extend)
-A2	DI6	IN	-B7	z_home	NO	Z home reference
-A2	DI7	IN	-B8	product_drop_parking	NO	Product parking / drop a
-A2	DI8	IN	-B9	product_drop_sensor	NO	Product drop verify (E3
-A2	DI9	IN	-B10	product_pickup_sensor	NO	Pickup door sensor (E3
-A2	DI10	IN	-K1	estop_feedback	1 = K1 released	K1 aux NC 21-22 feedb
-A2	DO0	OUT	-P1	ready_lamp	1 = on	Green pilot light READY
-A2	DO1	OUT	-P2	moving_lamp	1 = on	Yellow pilot light MOVIN
-A2	DO2	OUT	-P3/-P4	alarm_lamp	1 = on	Red pilot light + buzzer
-A2	DO3	OUT	-Y1	dispense_actuator	1 = open	Dispense gate solenoid
-A3	PA8 TIM1_CH1	OUT	-A4:IN1	STEP_X	pulse	X step, 10 Hz - 50 kHz
-A3	PB0	OUT	-A4:IN2	DIR_X		X direction
-A3	PA9 TIM1_CH2	OUT	-A4:IN3	STEP_Y	pulse	Y step
-A3	PB1	OUT	-A4:IN4	DIR_Y		Y direction
-A3	PA5 TIM2_CH1	OUT	-A4:IN5	STEP_Z	pulse	Z step (LD2 disabled)
-A3	PB2	OUT	-A4:IN6	DIR_Z		Z direction

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

ราย

	5	6	7	8

cription
X alarm
Y alarm
X in-position
Y in-position
y contactor enable (via -S0)
ary output 1
ary output 2
ary output 3
limit / home
x limit
limit (drop level)
x limit
limit (retract)
x limit (extend)
ne reference
uct parking / drop align
uct drop verify (E3Z)
p door sensor (E3Z)
ux NC 21-22 feedback
n pilot light READY
w pilot light MOVING
ilot light + buzzer ALARM
nse gate solenoid
o, 10 Hz - 50 kHz
ction
o
ction
o (LD2 disabled)
ction

I/O LIST

รายการอินพุต / เอาต์พุต

DOC. NO. **AUTO-Parts-E01**	DRAWN PAPON	SHEET **19** of 22
REVISION **4.0**	CHECKED	
DATE **2026-09-26**	APPROVED	

1	2	3	4

Wire no. = <sheet><nn>. Sheet ref = sheet.column where the wire is drawn.

Wire	From	To	Colour	mm2	Type	Potential / signal
401	-X0:L	-Q1:1	BN	2.5	H07V-K	L
402	-X0:N	-Q1:3	BU	2.5	H07V-K	N
403	-X0:PE	-PE	GNYE	2.5	H07V-K	PE
404	-Q1:2	-Z1:L	BN	2.5	H07V-K	L
405	-Q1:4	-Z1:N	BU	2.5	H07V-K	N
406	-Q1:2	-F1:L	BN	2.5	H07V-K	L
407	-Q1:4	-F1:N	BU	2.5	H07V-K	N
408	-F1:PE	-PE	GNYE	4.0	H07V-K	PE
409	-Z1:PE	-PE	GNYE	2.5	H07V-K	PE
410	-Z1:L'	-X1:1	BN	2.5	H07V-K	L'
411	-Z1:N'	-X1:4	BU	2.5	H07V-K	N'
412	-X1:1	-F11:1	BN	1.5	H07V-K	L'
413	-F11:2	-T1:L	BN	1.5	H07V-K	L'
414	-X1:4	-T1:N	BU	1.5	H07V-K	N'
415	-PE	-T1:PE	GNYE	1.5	H07V-K	PE
416	-X1:2	-F12:1	BN	1.5	H07V-K	L'
417	-F12:2	-T2:L	BN	1.5	H07V-K	L'
418	-X1:5	-T2:N	BU	1.5	H07V-K	N'
419	-PE	-T2:PE	GNYE	1.5	H07V-K	PE
420	-X1:3	-F13:1	BN	1.5	H07V-K	L'
421	-F13:2	-T3:L	BN	1.5	H07V-K	L'
422	-X1:6	-T3:N	BU	1.5	H07V-K	N'
501	-T1:+V	-K1:1	RD	2.5	H07V-K	+60V
502	-T1:-V	-K1:3	BK	2.5	H07V-K	0V60
503	-K1:2	-F21:1	RD	2.5	H07V-K	+60V SW
504	-F21:1	-F22:1	RD	2.5	H07V-K	+60V SW
505	-F21:2	-U1:V+	RD	1.5	H07V-K	+60V X
506	-F22:2	-U2:V+	RD	1.5	H07V-K	+60V Y
507	-K1:4	-U1:V-	BK	1.5	H07V-K	0V60 SW
508	-K1:4	-U2:V-	BK	1.5	H07V-K	0V60 SW
509	-T2:+V	-F31:1	DBU	1.5	H07V-K	+24V
510	-T2:-V	-X2:9	DBU/WH	1.5	H07V-K	M
511	-F31:2	-X2:1	DBU	1.0	H07V-K	L+1
512	-F34:2	-X2:5	DBU	1.0	H07V-K	L+2
513	-F32:2	-X2:7	DBU	0.75	H07V-K	L+P
514	-F33:2	-K1:5	DBU	1.0	H07V-K	+24V Z
515	-K1:6	-U3:+V	DBU	1.0	H07V-K	+24V Z SW
516	-X2:10	-U3:GND	DBU/WH	1.0	H07V-K	M
517	-X2:9	-PE	GNYE	1.5	H07V-K	PE
601	-X2:5	-A1:DO_V+	DBU	0.75	H07V-K	L+2
602	-A1:DO0	-S0:11	DBU	0.75	Cable core	K1 enable
603	-S0:12	-K1:A1	DBU	0.75	Cable core	K1 coil
604	-K1:A2	-X2:10	DBU/WH	0.75	H07V-K	M
605	-X2:6	-K1:21	DBU	0.5	H07V-K	L+2
606	-K1:22	-A2:DI10	DBU	0.5	H07V-K	estop_feedback
701	-X2:1	-A1:24V	DBU	0.75	H07V-K	L+1

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

WI

	Cable	Sheet ref	Remark
		04.1	
		04.1	
		04.1	
		04.1	
		04.1	
		04.2	Twin ferrule at -Q1:2
		04.2	Twin ferrule at -Q1:4
		04.2	Shortest route, < 0.5 m
		04.1	
		04.1	
		04.1	
		04.3	
		04.3	
		04.3	
		04.3	
		04.5	
		04.5	
		04.5	
		04.5	
		04.7	
		04.7	
		04.7	-T3 class II, no PE
		05.1	
		05.1	
		05.2	
		05.3	Bridge
		05.2	
		05.3	
		05.2	Twin ferrule at -K1:4
		05.3	Twin ferrule at -K1:4
		05.5	-F31..-F34 inputs bridged by comb jumper
		05.4	
		05.5	
		05.6	
		05.6	
		05.7	
		05.7	
		05.7	
		05.4	Single-point earthing of M (PELV)
		06.2	
	-W601	06.2	
	-W601	06.2	
		06.2	
		06.3	
		06.3, 12.8	1 = K1 released
		07.6	

WIRE LIST (1/3)

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DOC. NO. **AUTO-Parts-E01**	DRAWN PAPON
REVISION **4.0**	CHECKED
DATE **2026-09-26**	APPROVED

SHEET **20** of 22

1	2	3	4

Wire no. = <sheet><nn>. Sheet ref = sheet.column where the wire is drawn.

Wire	From	To	Colour	mm2	Type	Potential / signal
702	-X2:11	-A1:0V	DBU/WH	0.75	H07V-K	M
703	-X2:2	-U1:ALM+	DBU	0.5	H07V-K	L+1
704	-U1:ALM+	-U1:PEND+	DBU	0.5	H07V-K	L+1
705	-U1:ALM-	-A1:DI0	DBU	0.5	H07V-K	X_DRIVE_ALM
706	-U1:PEND-	-A1:DI2	DBU	0.5	H07V-K	X_PEND
707	-X2:2	-U2:ALM+	DBU	0.5	H07V-K	L+1
708	-U2:ALM+	-U2:PEND+	DBU	0.5	H07V-K	L+1
709	-U2:ALM-	-A1:DI1	DBU	0.5	H07V-K	Y_DRIVE_ALM
710	-U2:PEND-	-A1:DI3	DBU	0.5	H07V-K	Y_PEND
711	-A1:DI_COM	-X2:11	DBU/WH	0.5	H07V-K	M
901	-A3:PA8 (CN10-23)	-A4:IN1	(maker)	0.25	Jumper lead	STEP_X
902	-A3:PB0 (CN7-34)	-A4:IN2	(maker)	0.25	Jumper lead	DIR_X
903	-A3:PA9 (CN10-21)	-A4:IN3	(maker)	0.25	Jumper lead	STEP_Y
904	-A3:PB1 (CN10-24)	-A4:IN4	(maker)	0.25	Jumper lead	DIR_Y
905	-A3:PA5 (CN10-11)	-A4:IN5	(maker)	0.25	Jumper lead	STEP_Z
906	-A3:PB2 (CN10-22)	-A4:IN6	(maker)	0.25	Jumper lead	DIR_Z
907	-A3:GND (CN10-20)	-A4:GND	(maker)	0.25	Jumper lead	GND
908	-A4:GND	-X2:12	DBU/WH	0.5	H07V-K	M
909	-A4:OUT1	-U1:PUL-	DBU	0.5	Twisted pair	STEP_X
910	-A4:OUT2	-U1:DIR-	DBU	0.5	Twisted pair	DIR_X
911	-A4:OUT3	-U2:PUL-	DBU	0.5	Twisted pair	STEP_Y
912	-A4:OUT4	-U2:DIR-	DBU	0.5	Twisted pair	DIR_Y
913	-A4:OUT5	-U3:PUL-	DBU	0.5	Twisted pair	STEP_Z
914	-A4:OUT6	-U3:DIR-	DBU	0.5	Twisted pair	DIR_Z
1001	-U1:A+	-M1:A+	RD	1.0	Motor cable core	Phase A+
1002	-U1:A-	-M1:A-	BU	1.0	Motor cable core	Phase A-
1003	-U1:B+	-M1:B+	BK	1.0	Motor cable core	Phase B+
1004	-U1:B-	-M1:B-	GN	1.0	Motor cable core	Phase B-
1011	-X2:7	-U1:PUL+	DBU	0.5	H07V-K	L+P
1012	-U1:PUL+	-U1:DIR+	DBU	0.5	H07V-K	L+P
1021	-U2:A+	-M2:A+	RD	1.0	Motor cable core	Phase A+
1022	-U2:A-	-M2:A-	BU	1.0	Motor cable core	Phase A-
1023	-U2:B+	-M2:B+	BK	1.0	Motor cable core	Phase B+
1024	-U2:B-	-M2:B-	GN	1.0	Motor cable core	Phase B-
1031	-X2:8	-U2:PUL+	DBU	0.5	H07V-K	L+P
1032	-U2:PUL+	-U2:DIR+	DBU	0.5	H07V-K	L+P
1101	-U3:A+	-M3:A+	RD	0.5	Motor cable core	Phase A+
1102	-U3:A-	-M3:A-	BU	0.5	Motor cable core	Phase A-
1103	-U3:B+	-M3:B+	BK	0.5	Motor cable core	Phase B+
1104	-U3:B-	-M3:B-	GN	0.5	Motor cable core	Phase B-
1111	-X2:8	-U3:PUL+	DBU	0.5	H07V-K	L+P
1112	-U3:PUL+	-U3:DIR+	DBU	0.5	H07V-K	L+P
1201	-X2:3	-X3:11	DBU	0.75	H07V-K	L+1
1202	-X2:12	-X3:21	DBU/WH	0.75	H07V-K	M
1203	-X2:3	-A2:24V	DBU	0.75	H07V-K	L+1
1204	-X2:13	-A2:0V	DBU/WH	0.75	H07V-K	M

PAPON

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PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

WI

	Cable	Sheet ref	Remark
		07.6	
		07.3	
		07.3	Jumper
		07.3	
		07.3	
		07.4	
		07.5	Jumper
		07.4	
		07.5	
		07.3	
	-W901	09.3	TIM1_CH1
	-W901	09.3	
	-W901	09.3	TIM1_CH2
	-W901	09.3	
	-W901	09.3	TIM2_CH1, LD2 disabled
	-W901	09.3	
	-W901	09.3	
		09.4	0 V reference of NMOS sources
		09.5, 10.1	Twist with 1011
		09.5, 10.1	Twist with 1012
		09.5, 10.5	Twist with 1031
		09.5, 10.5	Twist with 1032
		09.5, 11.2	Twist with 1111
		09.5, 11.2	Twist with 1112
	-W1001	10.3	
	-W1001	10.3	
	-W1001	10.3	
	-W1001	10.3	
		10.1	
		10.2	Jumper
	-W1021	10.7	
	-W1021	10.7	
	-W1021	10.7	
	-W1021	10.7	
		10.5	
		10.6	Jumper
	-W1101	11.4	
	-W1101	11.4	
	-W1101	11.4	
	-W1101	11.4	
		11.2	
		11.2	Jumper
		12.1	Sensor supply, X3:11-20 bridged
		12.1	Sensor 0 V, X3:21-30 bridged
		12.1	
		12.2	

WIRE LIST (2/3)

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SHEET **21** of 22

1	2	3	4

Wire no. = <sheet><nn>. Sheet ref = sheet.column where the wire is drawn.

Wire	From	To	Colour	mm2	Type	Potential / signal
1205	-A2:0V	-A2:S/S	DBU/WH	0.5	H07V-K	M
1211	-X3:1	-A2:DI0	DBU	0.5	H07V-K	x_head_limit
1212	-X3:2	-A2:DI1	DBU	0.5	H07V-K	x_tail_limit
1213	-X3:3	-A2:DI2	DBU	0.5	H07V-K	y_head_limit
1214	-X3:4	-A2:DI3	DBU	0.5	H07V-K	y_tail_limit
1215	-X3:5	-A2:DI4	DBU	0.5	H07V-K	z_head_limit
1216	-X3:6	-A2:DI5	DBU	0.5	H07V-K	z_tail_limit
1217	-X3:7	-A2:DI6	DBU	0.5	H07V-K	z_home
1218	-X3:8	-A2:DI7	DBU	0.5	H07V-K	product_drop_parking
1219	-X3:9	-A2:DI8	DBU	0.5	H07V-K	product_drop_sensor
1220	-X3:10	-A2:DI9	DBU	0.5	H07V-K	product_pickup_sensor
1301	-X2:4	-A2:DO_COM	DBU	0.75	H07V-K	L+1
1302	-A2:DO0	-P1:X1	DBU	0.75	Cable core	ready_lamp
1303	-A2:DO1	-P2:X1	DBU	0.75	Cable core	moving_lamp
1304	-A2:DO2	-P3:X1	DBU	0.75	Cable core	alarm_lamp
1305	-P3:X1	-P4:+	DBU	0.75	H07V-K	alarm_lamp
1306	-A2:DO3	-Y1:A1	DBU	0.75	Cable core	dispense_actuator
1307	-P1:X2	-P2:X2	DBU/WH	0.75	H07V-K	M
1308	-P2:X2	-P3:X2	DBU/WH	0.75	H07V-K	M
1309	-P3:X2	-P4:-	DBU/WH	0.75	H07V-K	M
1310	-P4:-	-X2:13	DBU/WH	0.75	Cable core	M
1311	-Y1:A2	-X2:14	DBU/WH	0.75	Cable core	M

PAPON

DRAFT FOR REVIEW

PROJECT

AUTO ELECTRONIC PARTS BOX

Auto Electronic Parts Box

Electrical schematic | IEC 61082 / 81346

SHEET TITLE

WI

	Cable	Sheet ref	Remark
		12.2	Jumper: inputs for PNP sensors
		12.1	
		12.2	
		12.2	
		12.3	
		12.4	
		12.4	
		12.5	
		12.6	
		12.6	
		12.7	
		13.1	
	-W1301	13.2	
	-W1301	13.3	
	-W1301	13.5	
		13.5	Jumper on door
	-W1302	13.6	
		13.2	Jumper on door
		13.3	Jumper on door
		13.5	Jumper on door
	-W1301	13.5	
	-W1302	13.6	

WIRE LIST (3/3)

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